



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 - Electricity Markets and Energy System Planning, Specialisation Course

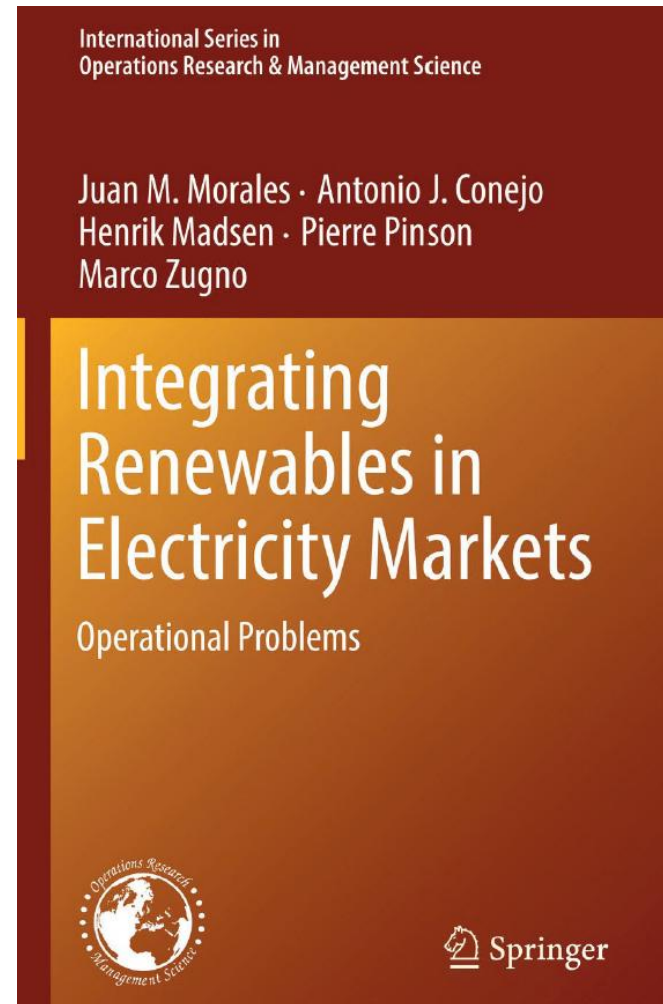
Risk Management for Multi-Market Bidding

Hossein Farahmand
Professor, IEL, NTNU
Autumn 2026



Reference

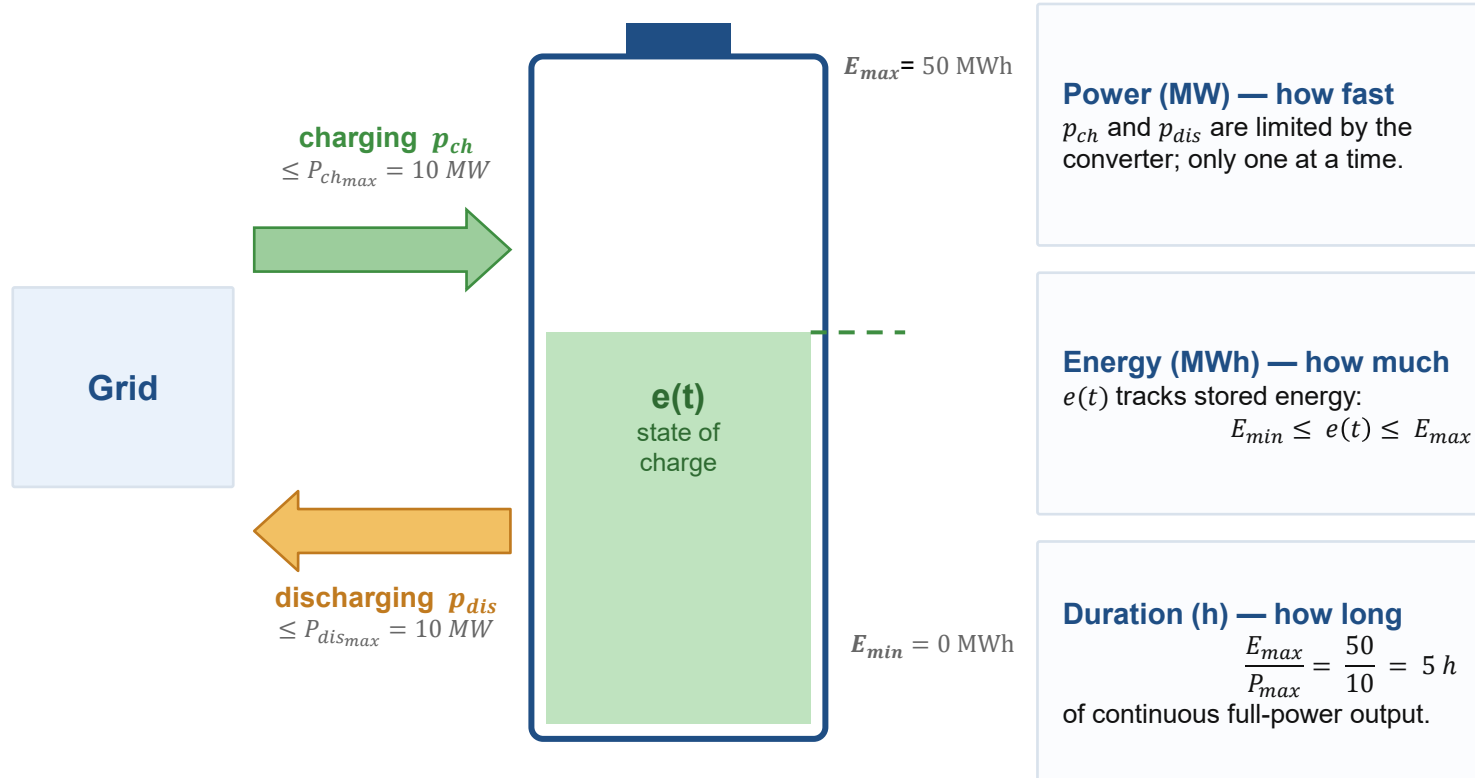
Juan M. Morales, Antonio J. Conejo,
Henrik Madsen, Pierre Pinson, Marco
Zugno, “Integrating Renewables in
Electricity Markets”, 2014, Volume
205, ISBN : 978-1-4614-9410-2



Overview

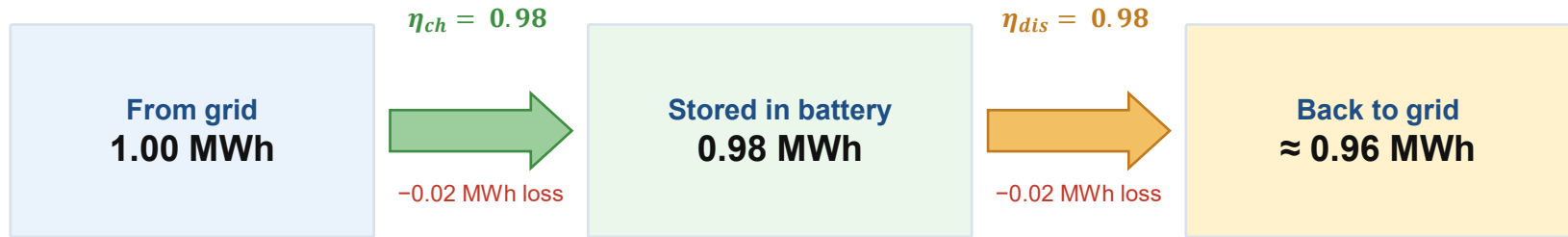
- Battery modelling basics: power, state of charge, efficiency
- Price-taking offering strategy and non-anticipativity
- From expected profit to downside-risk management
- VaR and CVaR as tail-risk measures
- Risk-averse stochastic offering model
- Battery DA + mFRR multi-market example
- Validation and project-use discussion

Battery basics: power, energy and State of Charge (SOC)



Two ratings define a battery, 10 MW power and 50 MWh energy, and one state variable, $e(t)$, links every decision over time.

Efficiency: every cycle loses some energy



In the SOC equation — charging

1 MWh drawn adds $\eta_{ch} \cdot \Delta p_{ch} = 0.98$ MWh to the battery.

In the SOC equation — discharging

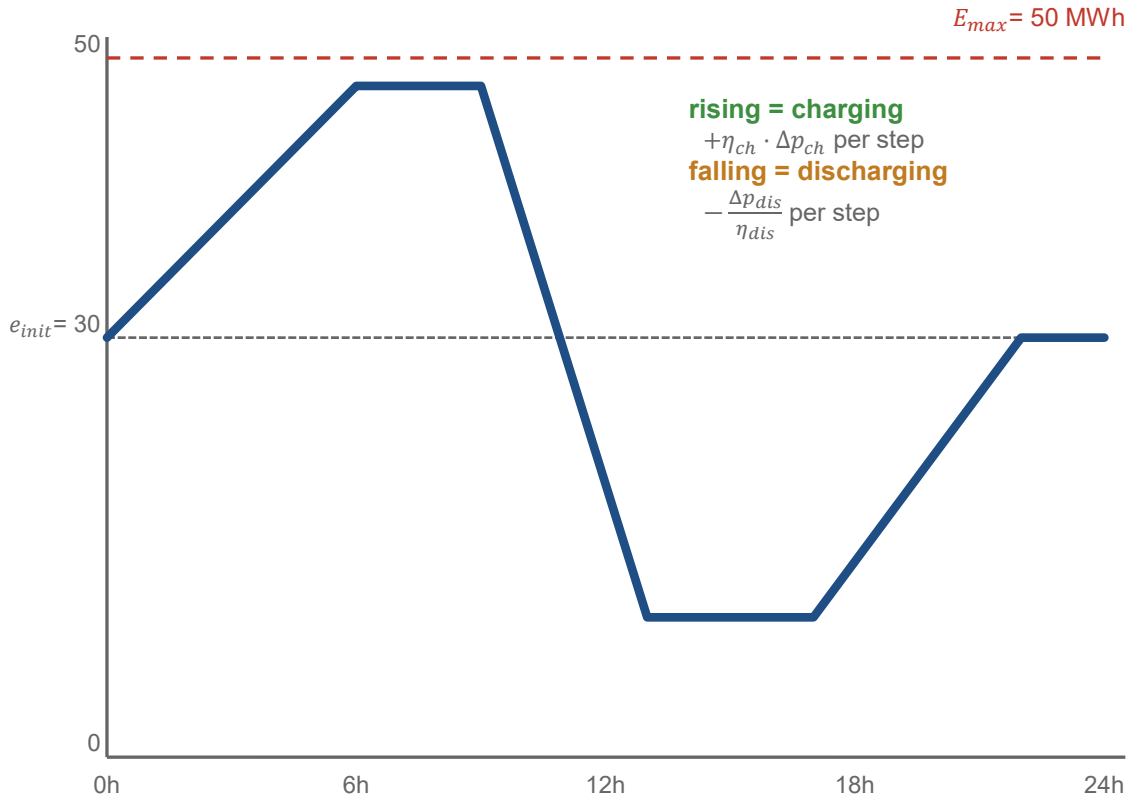
Delivering 1 MWh drains $\frac{\Delta p_{dis}}{\eta_{dis}} = 1.02$ MWh from the battery.

Round-trip efficiency $\eta_{ch} \cdot \eta_{dis} = 0.98 \times 0.98 \approx 0.96 \rightarrow \approx 4\%$ of every cycle is lost

Consequence for bidding: the price spread earned by a cycle (plus grid tariff) must beat the $\approx 4\%$ round-trip loss.

State of charge: the battery's memory

SOC $e(t)$ [MWh]



SOC bookkeeping

$$e[t+1] = e[t]$$

with $\Delta = 0.25 \text{ h}$ and
 $0 \leq e(t) \leq E_{max}$ at all times.

Slope limits: e can change by
at most $\pm \approx 2.5 \text{ MWh}$ per quarter
($10 \text{ MW} \times 0.25 \text{ h}$).

SOC couples all decisions in time and across markets: energy sold now is energy unavailable later.

Nordic battery reserve products

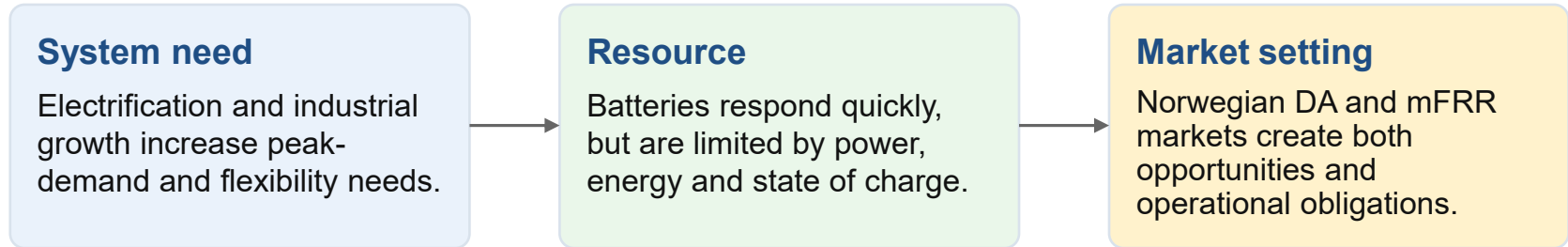
Activation time, endurance, direction, minimum bid and main battery constraints — Norway

Product	Activation time	Endurance	Direction	Min. bid	Main battery constraint
FFR	0.7, 1.0 or 1.3 s	5 or 30 s	Up	1 MW	Limited procurement hours and market volume
FCR-D	86% within 7.5 s	At least 20 min	Up/down	1 MW	Control tuning and counter-direction reservation
FCR-N	63% within 60 s; 95% within 3 min	1 h in each direction	Symmetric	0.1 MW	SoC management, energy availability and frequent activation
aFRR	Within 5 min	1 h	Up/down	1 MW	Sustained activation, communication requirements and energy availability
mFRR	12.5 min full activation	One or two quarters in EAM	Up/down	1 MW	Headroom, footroom, SoC management and energy deliverability

Source: Statnett / Entsoe — Norwegian reserve market specifications

Source: Nora Ingvild Gulbrandsen, "Improving Battery Storage Profitability through Value Stacking in Norway's Day-Ahead and mFRR Markets", Master thesis, 2026

Motivation from a Norwegian BESS value-stacking study



Why this matters for risk management:

- The same reserve bid can be profitable in one scenario and restrictive in another.
- A battery must keep enough energy for upward activation and enough free capacity for downward activation.
- Therefore, risk is not only financial: it is also feasibility risk under uncertain activation.

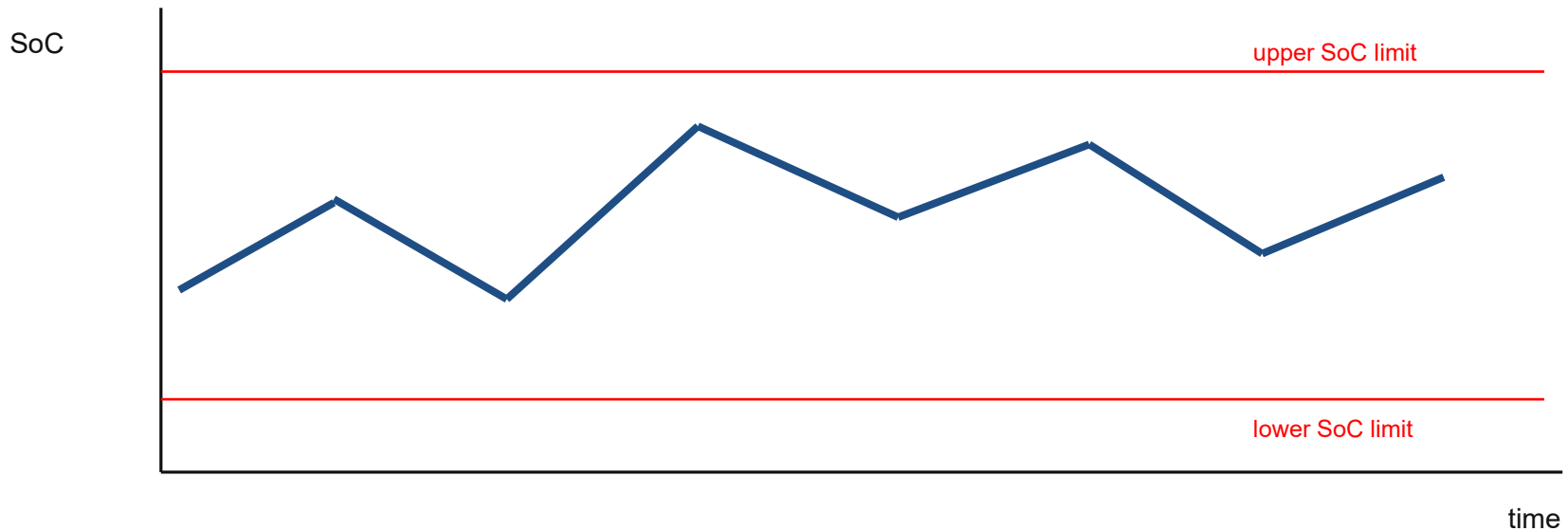
Battery bidding is constrained by state of charge

10 MW / 50 MWh example

5-hour nominal duration

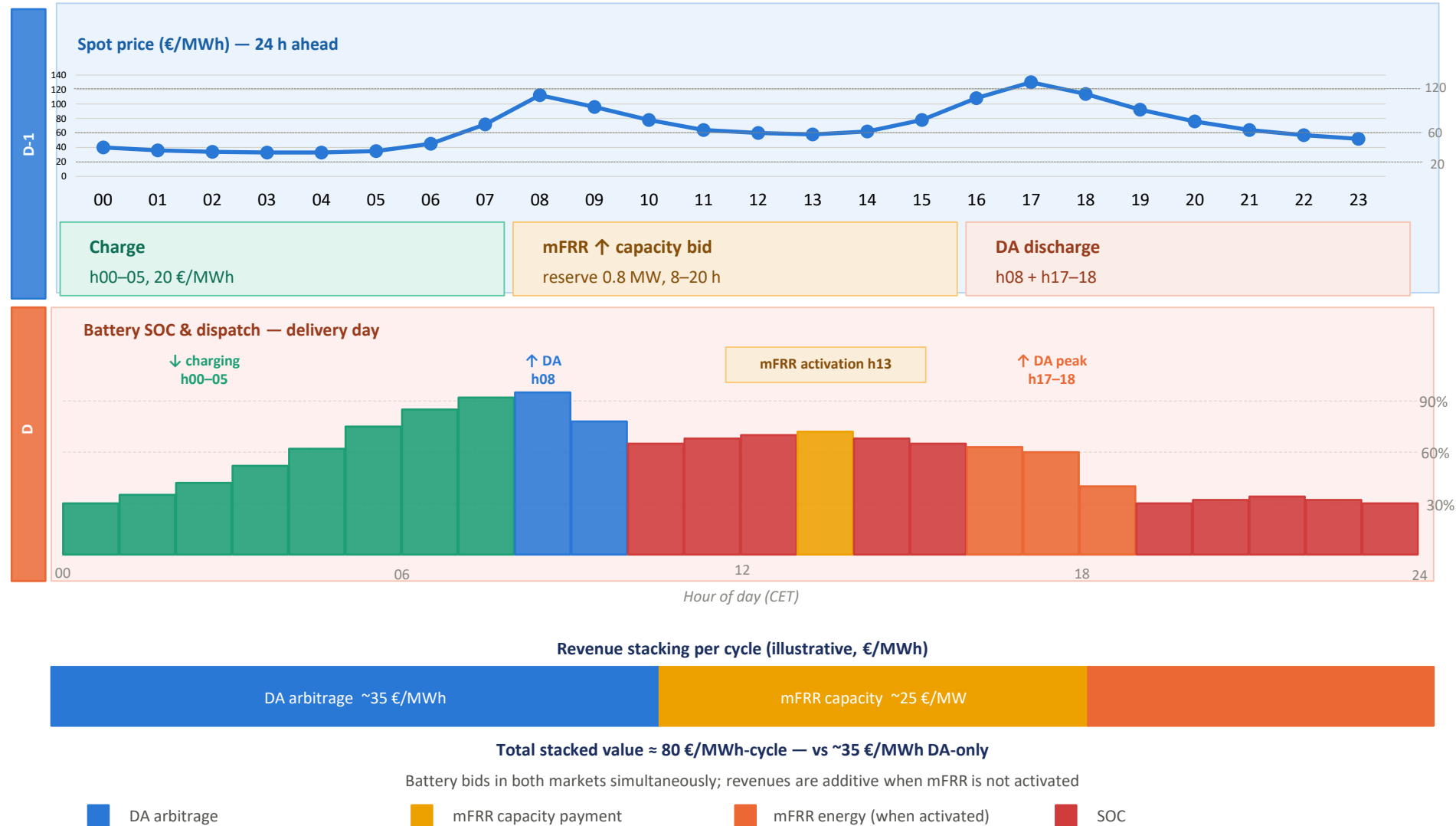
Reserve deliverability

Upward mFRR needs energy; downward mFRR needs free capacity



Value stacking is constrained: energy sold in one market cannot be freely sold again in another.

Battery value stacking — Nordic day-ahead + mFRR



Bidding-zone dependence: same model, different value



NO3

Lower DA price level and lower median mFRR activation prices, but important upward price spikes.

NO2

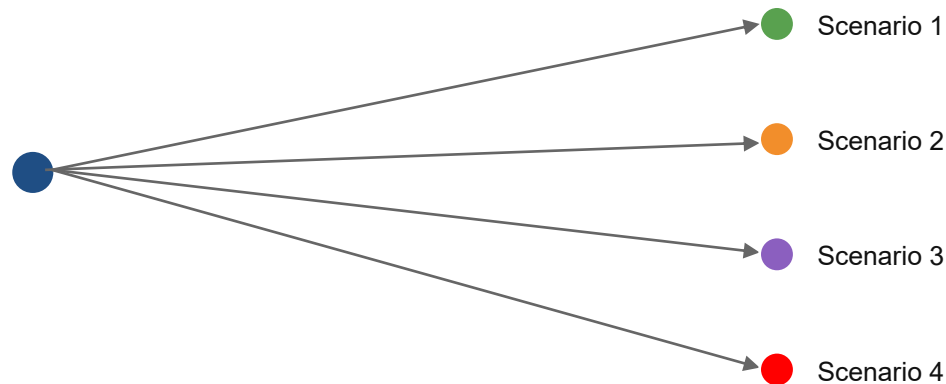
Higher price level and stronger value-stacking potential in the thesis case.

Multi-market bidding is not one universal answer. The optimal bid depends on local price distributions, activation tails, and SoC constraints.

Figure 2.4.1: Norwegian bidding zones (NO1–NC)

This is why scenario construction and out-of-sample testing matter.

Applied framework: two-stage planning and delivery



First-stage decisions are not scenario-indexed; second-stage recourse adapts to realized conditions.

Motivating result: value stacking is not automatically profitable

Zone	DA only	DA + mFRR capacity	Full DA + mFRR EAM
NO3	22.2 kEUR	16.6 kEUR	78.9 kEUR
NO2	305.9 kEUR	—	973.0 kEUR

Opportunity cost of reserves

In NO3, adding mFRR capacity alone reduced net profit because reserve commitments limited DA arbitrage.

Activation value

When voluntary mFRR activation was included, net profit increased strongly.

Risk-aware bidding asks: how much expected profit should we sacrifice to avoid bad tail outcomes and infeasible activation?

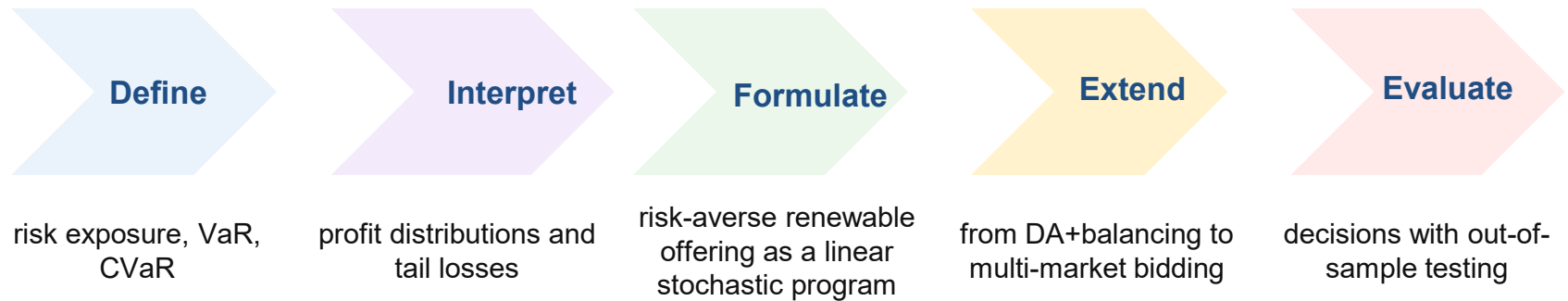
Source: Nora Ingvild Gulbrandsen, "Improving Battery Storage Profitability through Value Stacking in Norway's Day-Ahead and mFRR Markets", Master thesis, 2026

Where this lecture fits

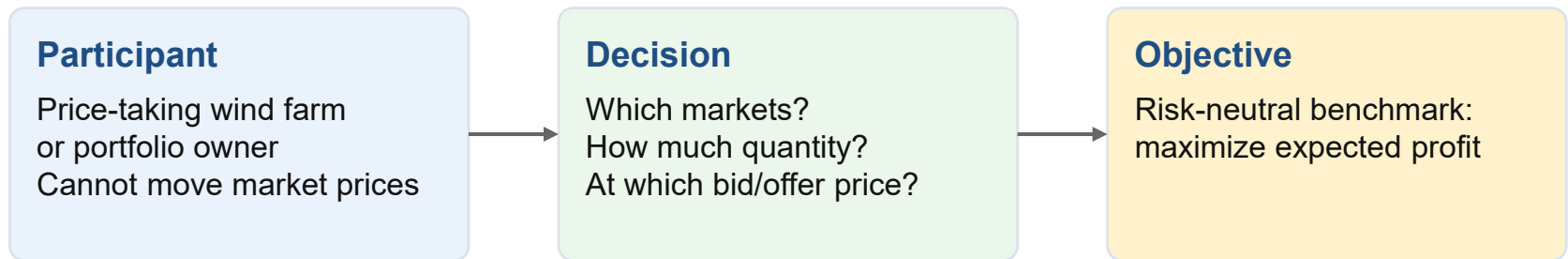
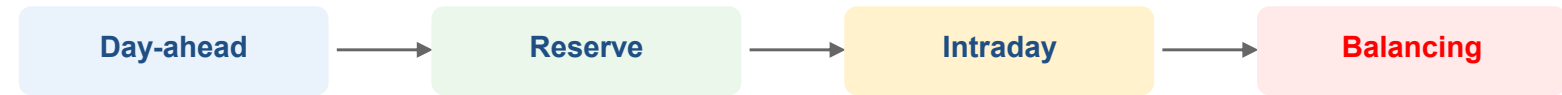


After L02: keep the two-stage scenario model, but replace "best in expectation" by "profitable enough in the bad tail".

Learning objectives

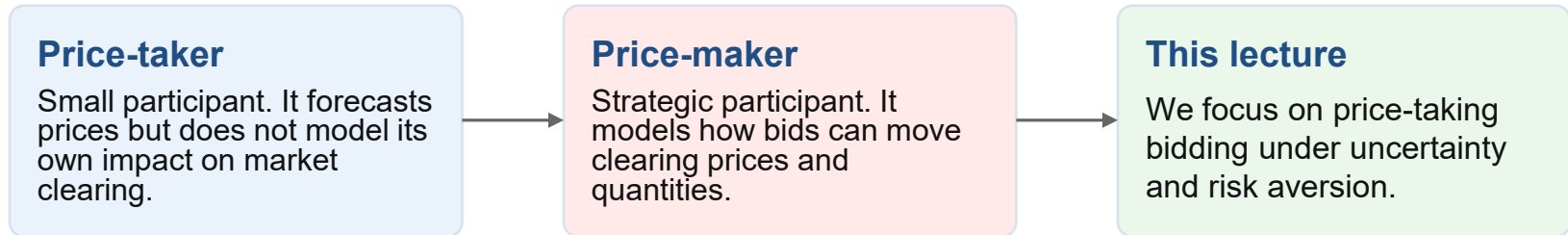


Recap: the offering strategy problem



Risk management asks: are we comfortable with the worst outcomes of the selected bid?

Price-taking assumption: what the bidder optimizes



For a price-taker, market prices are scenario inputs, not decision-dependent outcomes.

Decision problem viewpoint:

choose quantities → receive stochastic prices and stochastic physical outcomes → evaluate profit distribution

Why risk is unavoidable in renewable bidding

Volume risk

Wind/solar output differs from the DA forecast.

Price risk

DA, reserve, intraday and balancing prices are uncertain.

Activation risk

Reserve capacity can be activated with uncertain direction and volume.

Penalty risk

Imbalances can trigger costly settlement.

All sources are translated into scenario cash flow:

$$\text{Profit}[\omega] = \text{market revenues}[\omega] - \text{imbalance costs}[\omega] - \text{penalties}[\omega]$$

Risk-neutral benchmark: DA + balancing

Maximize expected profit:

$$\max_{p_t^{\text{DA}}, \delta_{t,\omega}^+, \delta_{t,\omega}^-} \sum_{\omega \in \Omega} \pi_{\omega} \text{profit}_{t,\omega}$$

Scenario profit:

$$\text{profit}_{t,\omega} = \lambda_{t,\omega}^{\text{DA}} p_t^{\text{DA}} + \lambda_{t,\omega}^{\text{up}} \delta_{t,\omega}^+ - \lambda_{t,\omega}^{\text{down}} \delta_{t,\omega}^- \quad \forall t \in \mathcal{T}, \omega \in \Omega,$$

Constraints:

$$0 \leq p_t^{\text{DA}} \leq P^{\text{nom}}, \forall t \in \mathcal{T},$$

$$\delta_{t,\omega} = p_{t,\omega}^{\text{real}} - p_t^{\text{DA}}, \quad \forall t \in \mathcal{T}, \omega \in \Omega,$$

$$\delta_{t,\omega} = \delta_{t,\omega}^+ - \delta_{t,\omega}^- \quad \forall t \in \mathcal{T}, \omega \in \Omega$$

$$\delta_{t,\omega}^+ \geq 0, \quad \delta_{t,\omega}^- \geq 0, \quad \forall t \in \mathcal{T}, \omega \in \Omega$$

Here-and-now

p_t^{DA} is chosen before the scenario is revealed.

Wait-and-see

Imbalance variables are scenario dependent.

Risk-neutral

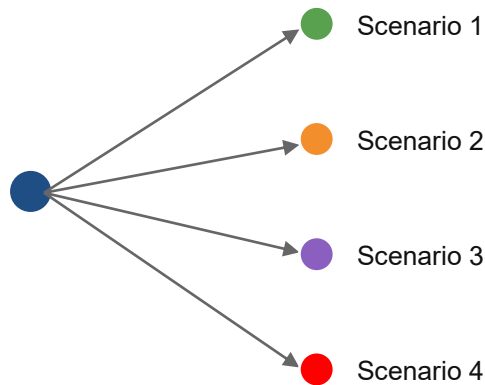
Only the probability-weighted mean matters.

Here, $\lambda_{t,\omega}^{\text{DA}}$, $\lambda_{t,\omega}^{\text{up}}$, and $\lambda_{t,\omega}^{\text{down}}$ represent the day-ahead, upward-imbalance, and downward-imbalance prices, respectively.

One bid - many possible real-time outcomes

Day-ahead stage

Balancing stage



Non-anticipativity:

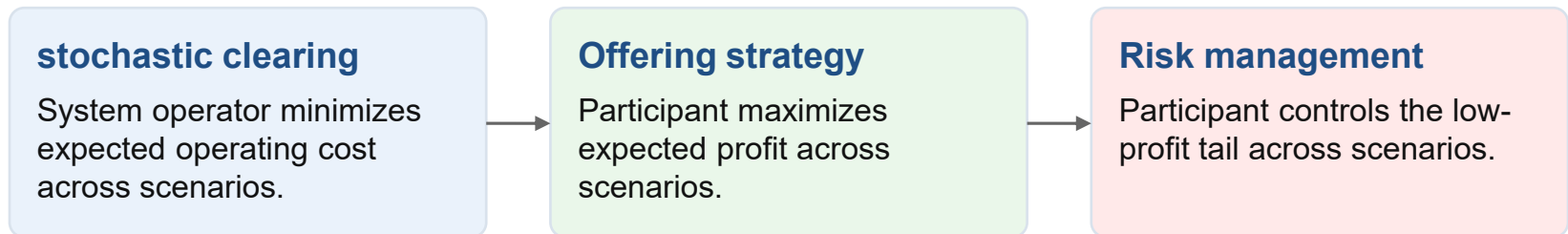
p_t^{DA} is the same
for every scenario w .

Recourse:

$\delta_{t,\omega}^+$ and $\delta_{t,\omega}^-$ adapt after uncertainty is
revealed.

Risk-aware bidding changes the first-stage bid by considering the tail of scenario profits.

Same two-stage structure - new economic question



Risk-neutral objective:

$$\max_x \mathbb{E}[\text{profit}(x)]$$

Optimizes the probability-weighted mean

Risk-aware objective:

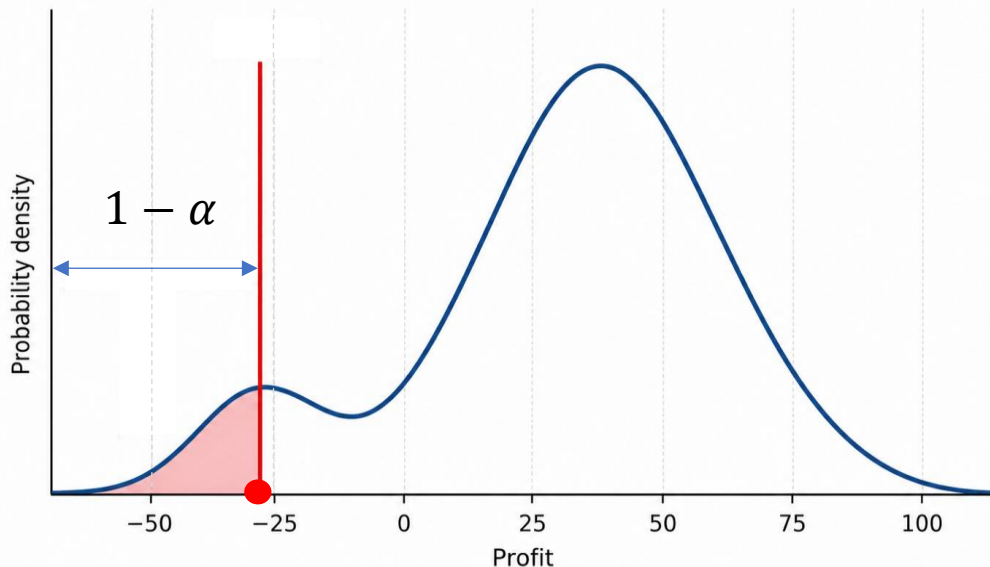
$$\max_x \mathbb{E}[\text{profit}(x)] + \text{tail protection}$$

Improves robustness against bad scenarios.

Expected profit alone can hide scenarios that threaten solvency, contracts, or project economics.

Value at Risk (VaR): where the downside tail begins

The Probability Density Function (PDF) of the Battery's profit might look as follows:



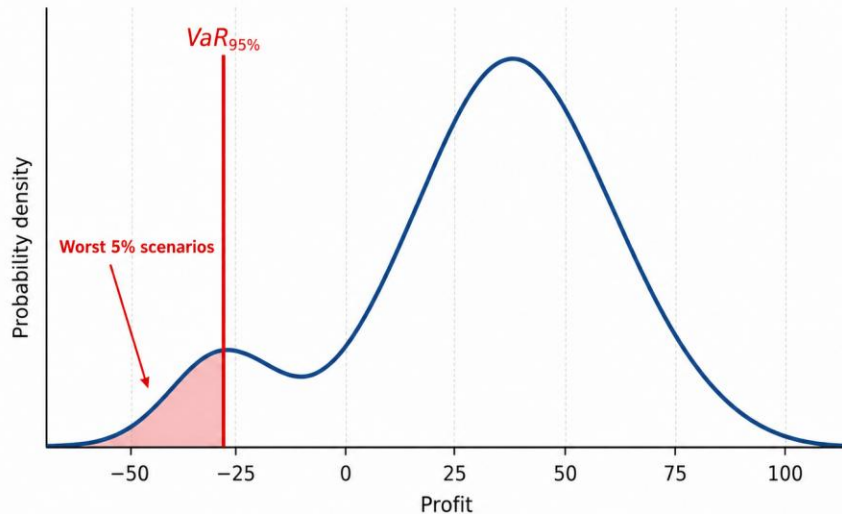
The **lower-profit tail** of the distribution, which the battery operator might be particularly concerned about, especially if it is a "fat" tail

$(1 - \alpha)$ -quantile of the distribution, where $0 \leq \alpha \leq 1$

VaR (Value-at-Risk) is the minimum profit such that the probability of the profit being lower than this value is less than or equal to $1 - \alpha$.

Value at Risk (VaR): where the downside tail begins

VaR is a profit threshold: only a small probability of scenarios falls below it.



Definition

$$\text{VaR}_{95\%}(\pi) = q_{5\%}(\pi)$$

$$\Pr(\pi \leq \text{VaR}_{95\%}) \approx 5\%$$

Interpretation for profit

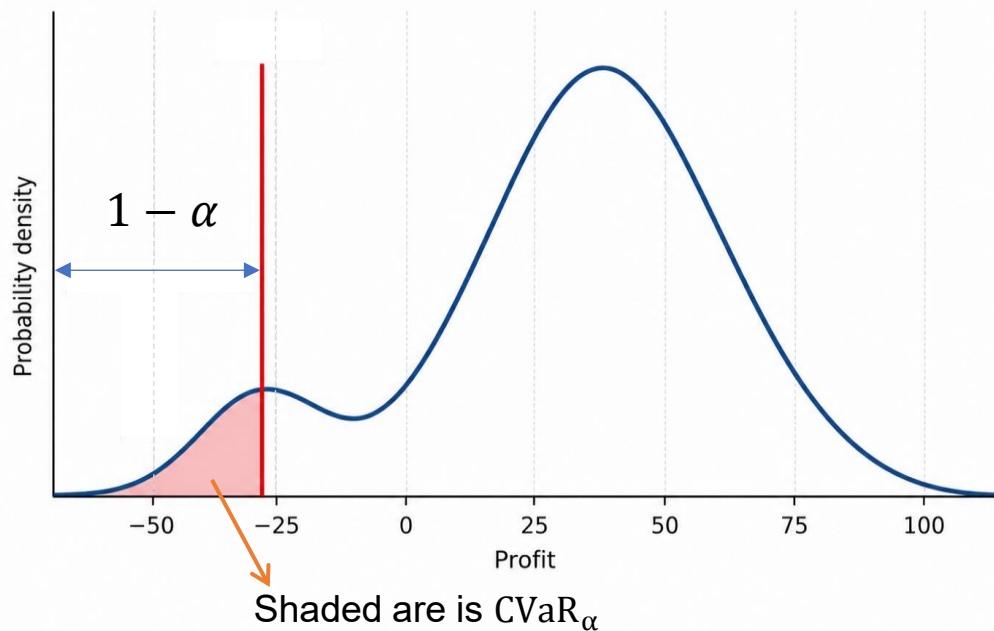
For **VaR_{95%}**, about 5% of scenarios have profit below the VaR threshold.

VaR tells us where the bad tail starts, but not how severe the losses are beyond it.

VaR identifies where the bad tail begins, but it does not measure how severe the outcomes beyond that threshold are.

Conditional value-at-risk (CVaR)

The Probability Density Function (PDF) of the Battery's profit might look as follows:



CVaR (Conditional Value-at-Risk) is the expected profit that is lower than the $(1-\alpha)$ -quantile of the profit distribution.

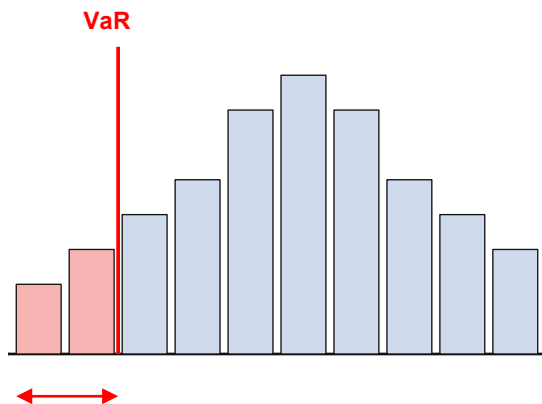
In other words, CVaR is the expected value in the tail region where profit is lower than the VaR.

Unlike VaR, CVaR is a key risk measure when the underlying distribution has a 'fat tail' (i.e., a larger tail region, or gray area, where extreme losses occur).

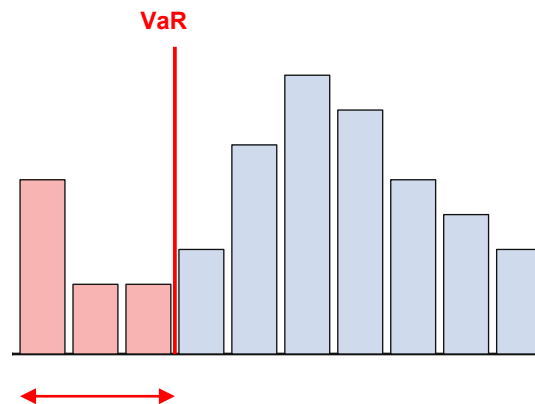
Expected profit can hide downside exposure

Two bids with similar expected profit

Bid A: stable tail



Bid B: dangerous tail

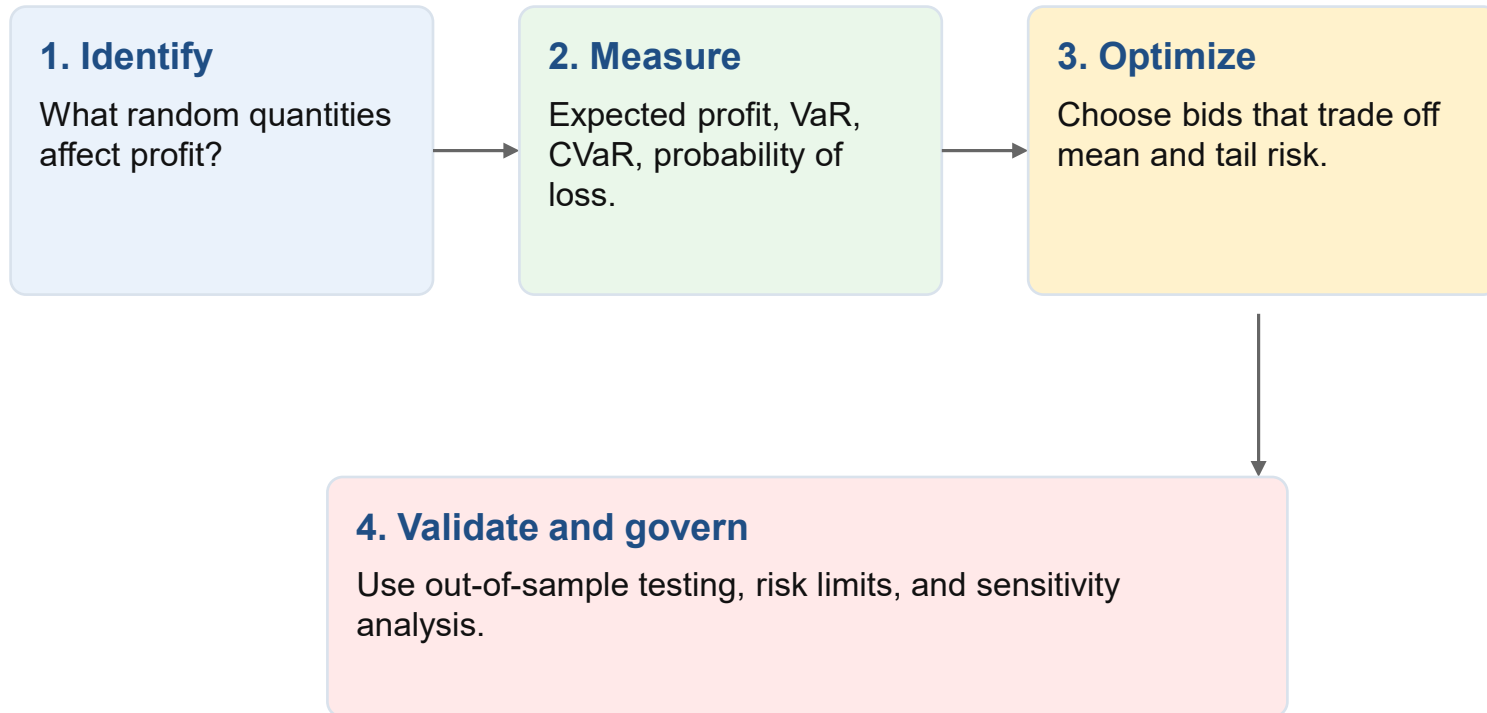


Two bids can have almost the same expected profit but very different risk profiles.

- **Bid A** has relatively mild losses in its worst scenarios.
- **Bid B** has more severe low-profit outcomes, even though its average profit may be similar or slightly higher.
- A risk-neutral model that maximizes only $\mathbb{E}[\text{profit}]$ may therefore select Bid B and ignore the possibility of unacceptable losses.

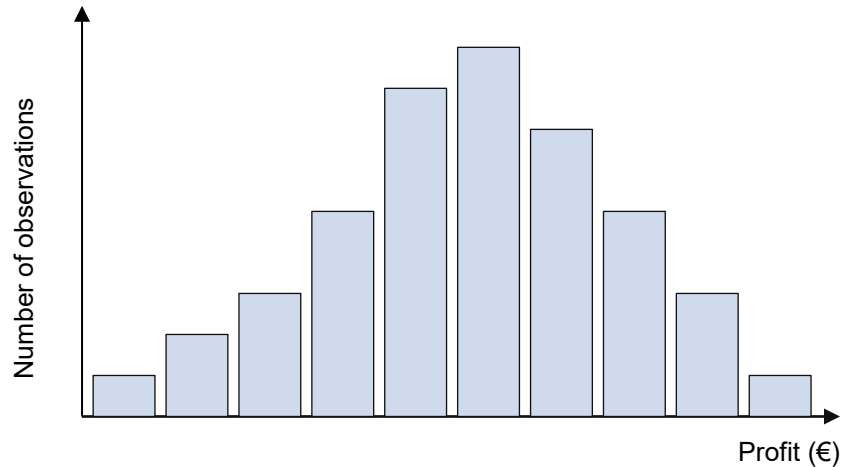
VaR identifies where the bad tail begins, but it does not measure how severe the outcomes beyond that threshold are. That is why CVaR is useful: it measures the average profit in the worst tail.

Risk management: control the shape of profit distribution



Conditional value-at-risk (CVaR):

Let's now consider a discrete distribution for multi-market profit:



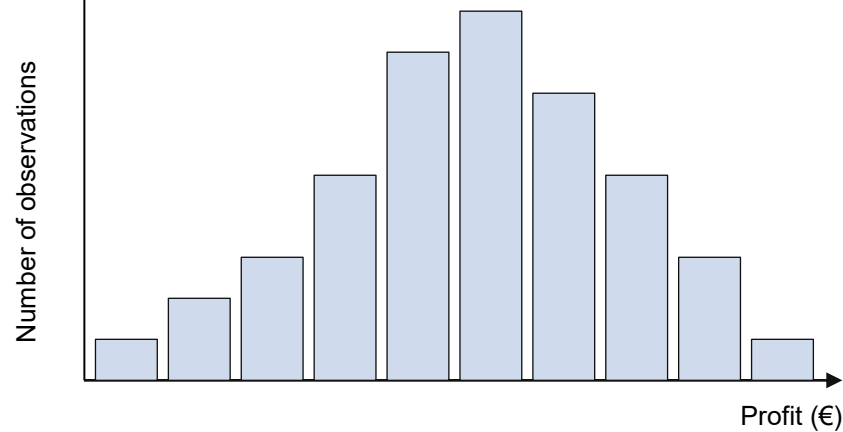
Maximization of the **expected profit**:

$$\underset{P^{DA}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}}{\text{maximize}} \quad \sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{DA}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})$$

$$\begin{aligned} \text{subject to} \quad & f(P^{DA}) \leq 0, \\ & g(P^{DA}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega. \end{aligned}$$

Conditional value-at-risk (CVaR):

Let's now consider a discrete distribution for multi-market profit:

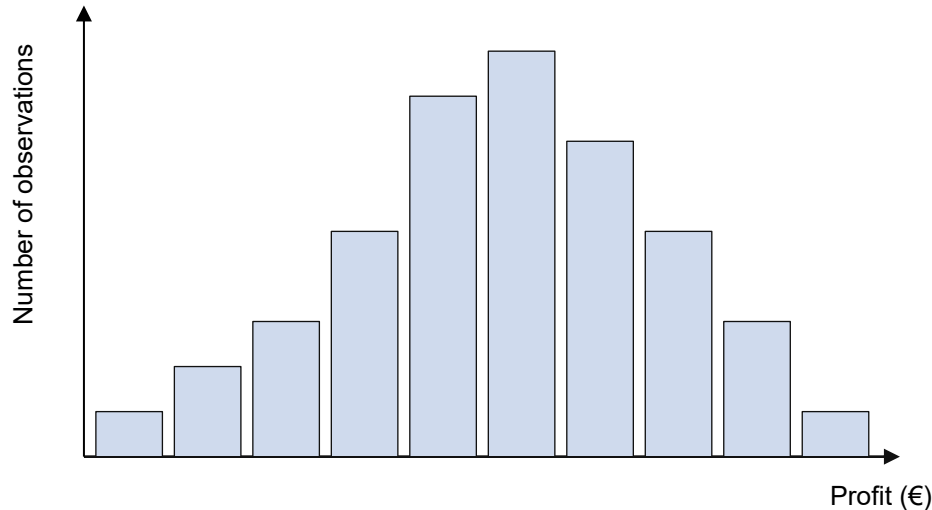


Maximization of the **expected profit**:

$$\begin{aligned}
 &\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}}{\text{maximize}} && \sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \\
 & && \text{Day-ahead offer (MW)} \quad \text{offer (MW)} \quad \text{Imbalances under scenario } \omega \text{ (MW)} \\
 &\text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega. \\
 & && \text{Technical constraints}
 \end{aligned}$$

Conditional value-at-risk (CVaR):

Let's now consider a discrete distribution for multi-market profit:

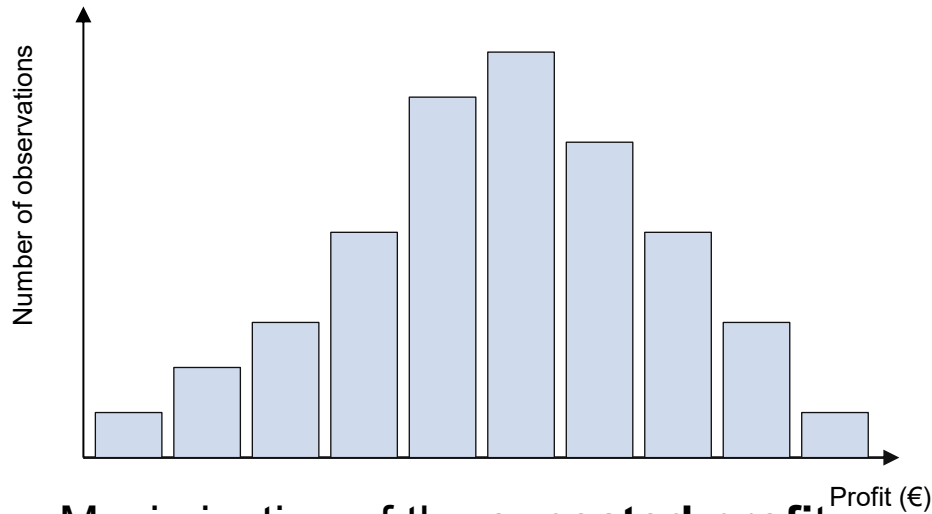


Maximization of the **expected profit**:

$$\begin{aligned}
 &\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}}{\text{maximize}} && \sum_{\omega} \underbrace{\pi_{\omega}}_{\substack{\text{Probability of scenario } \omega}} \underbrace{\text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})}_{\substack{\text{Profit function under scenario } \omega \text{ (€)}}} \\
 &\text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

Conditional value-at-risk (CVaR):

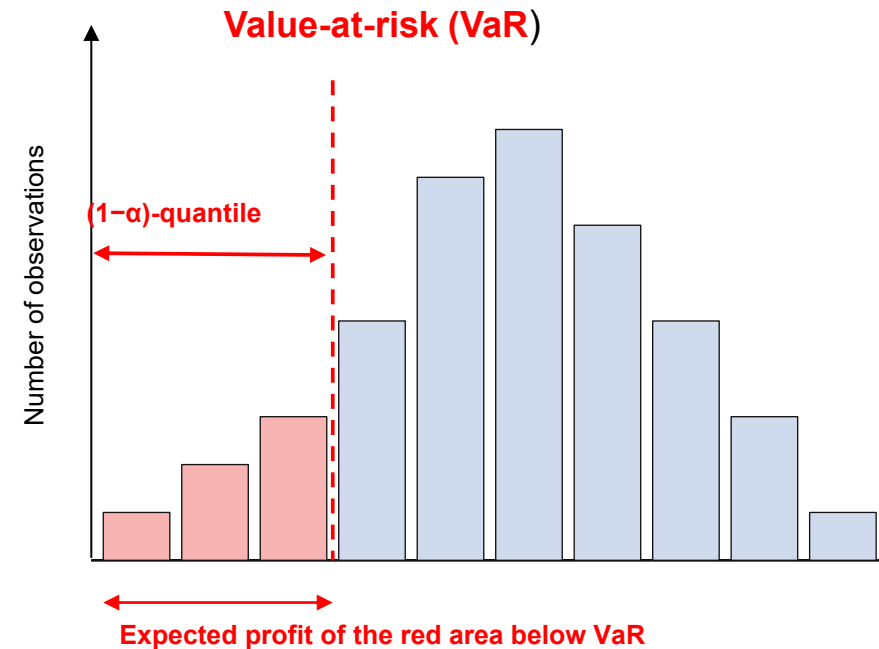
Let's now consider a discrete distribution for multi-market profit:



Maximization of the **expected profit**:

$$\begin{aligned}
 &\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}}{\text{maximize}} && \sum_{\omega} \underbrace{\pi_{\omega}}_{\substack{\text{Probability of scenario } \omega}} \underbrace{\text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})}_{\substack{\text{Profit function under scenario } \omega \text{ (€)}}} \\
 &\text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

Conditional value-at-risk (CVaR): average of the bad tail

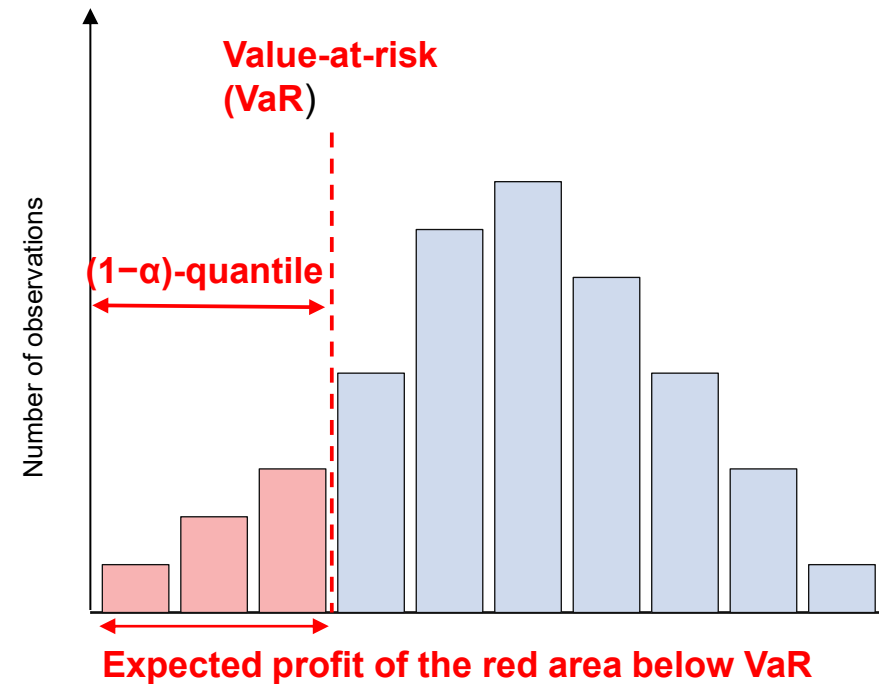


Maximization of the **CVaR**::

$$\begin{aligned}
 & \underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta}{\text{maximize}} && \underbrace{\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \max[0, \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})]}_{\text{The definition of CVaR (€) according to Rockafellar and Uryasev [1]}} \\
 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

[1] Rockafellar, R.T., Uryasev, S., "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, pp. 21–41, 2000

Conditional value-at-risk (CVaR): average of the bad tail



Maximization of the **CVaR**::

$$\text{maximize}_{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta} \quad \zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \max[0, \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})]$$

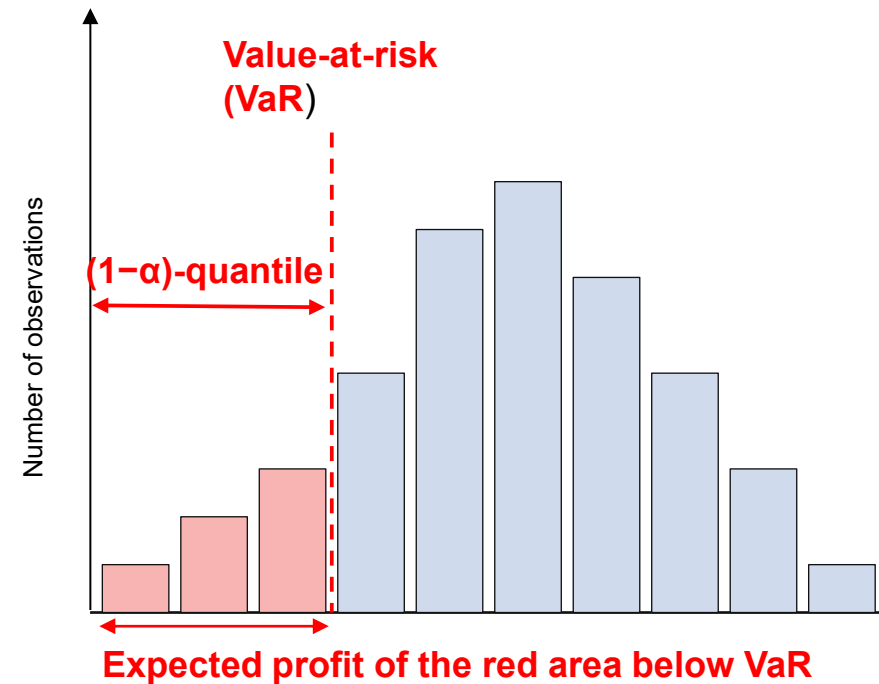
$$\text{subject to} \quad f(P^{\text{DA}}) \leq 0,$$

$$g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.$$

VaR (€) (it is a variable to be determined by the optimization problem)

Note that the optimal value of VaR is determined endogenously (it is not a parameter).

Conditional value-at-risk (CVaR): average of the bad tail

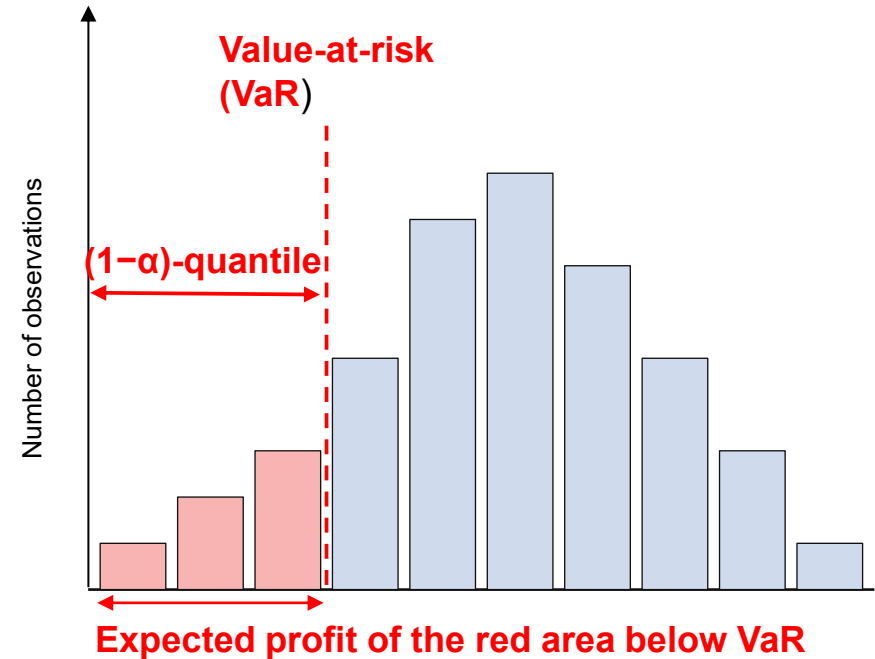


Maximization of the **CVaR**::

$$\begin{aligned}
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 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

How to reformulate the term **max[.,.]** in the objective function?

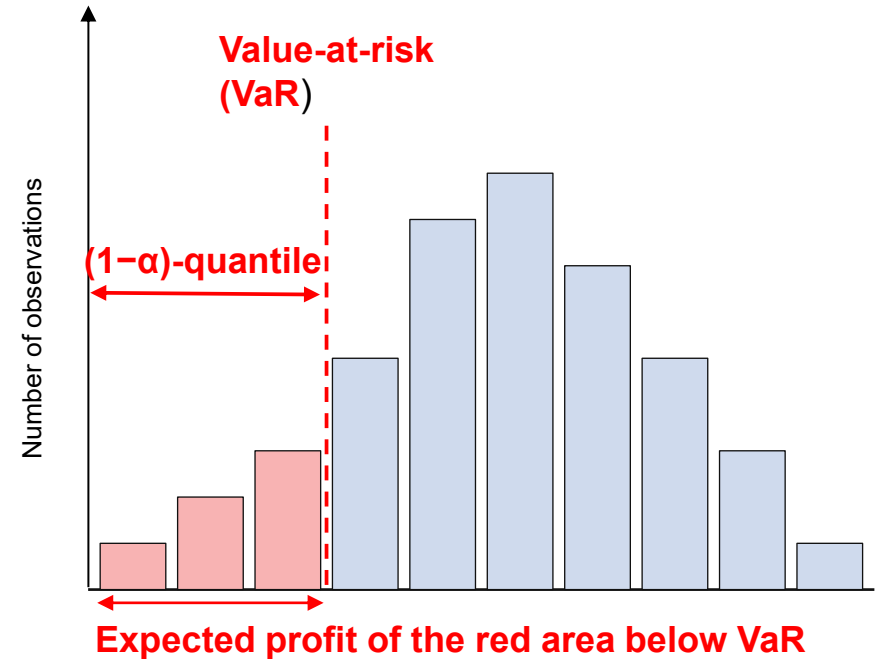
Conditional value-at-risk (CVaR): average of the bad tail



Reformulation of the CVaR-maximiation problem:

$$\begin{aligned}
 & \underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}}{\text{maximize}} && \zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \\
 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}), \quad \forall \omega.
 \end{aligned}$$

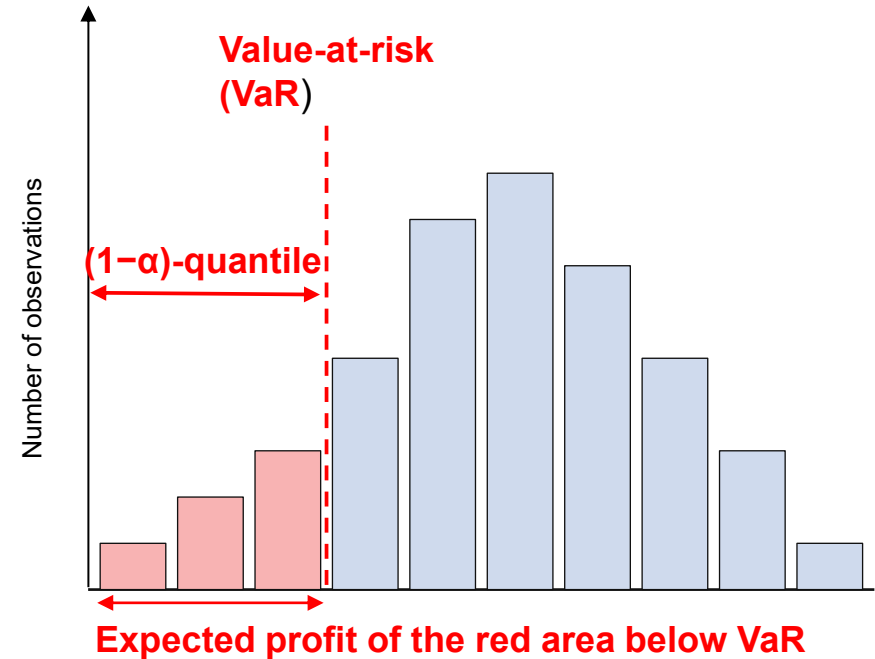
Conditional value-at-risk (CVaR): average of the bad tail



Reformulation of the CVaR-maximiation problem:

$$\begin{aligned}
 & \underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}}{\text{maximize}} && \zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \quad \text{An auxiliary variable (€)} \\
 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}), \quad \forall \omega.
 \end{aligned}$$

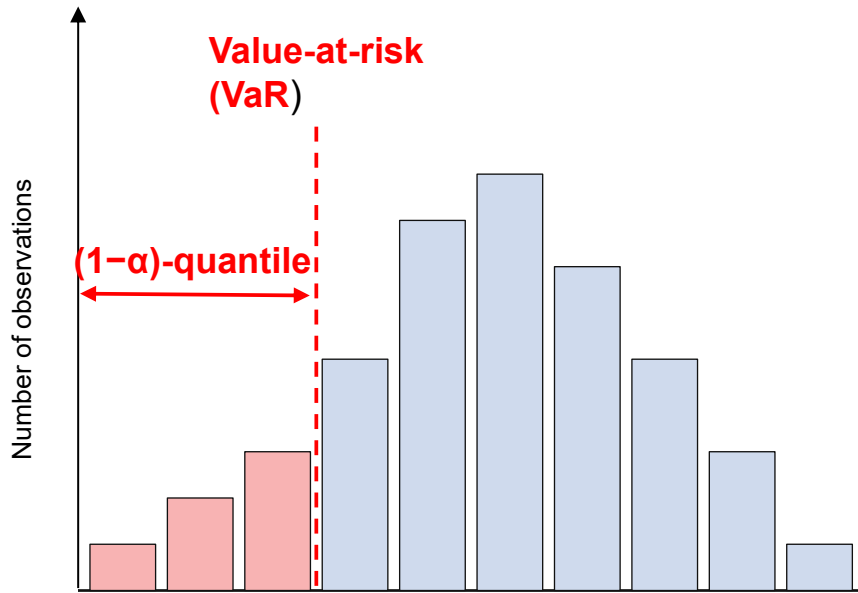
Conditional value-at-risk (CVaR): average of the bad tail



Reformulation of the CVaR-maximisation problem:

$$\begin{aligned}
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 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}), \quad \forall \omega.
 \end{aligned}$$

Expected profit vs. CVaR



Should we maximize expected profit for CVaR? Is there a way to consider both? **Yes, let's formulate a multiobjective optimization problem!**

Revised objective function:

Given non-negative weight (parameter)

$$\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} \quad \underbrace{\left[\sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \right]}_{\text{Expected profit (€)}} + \beta \underbrace{\left[\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \right]}_{\text{CVaR (€)}}.$$

Expected profit (€)

CVaR (€)

Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

$$\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} \left[\sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \right].$$

2) For $0 \leq \beta \leq 1$

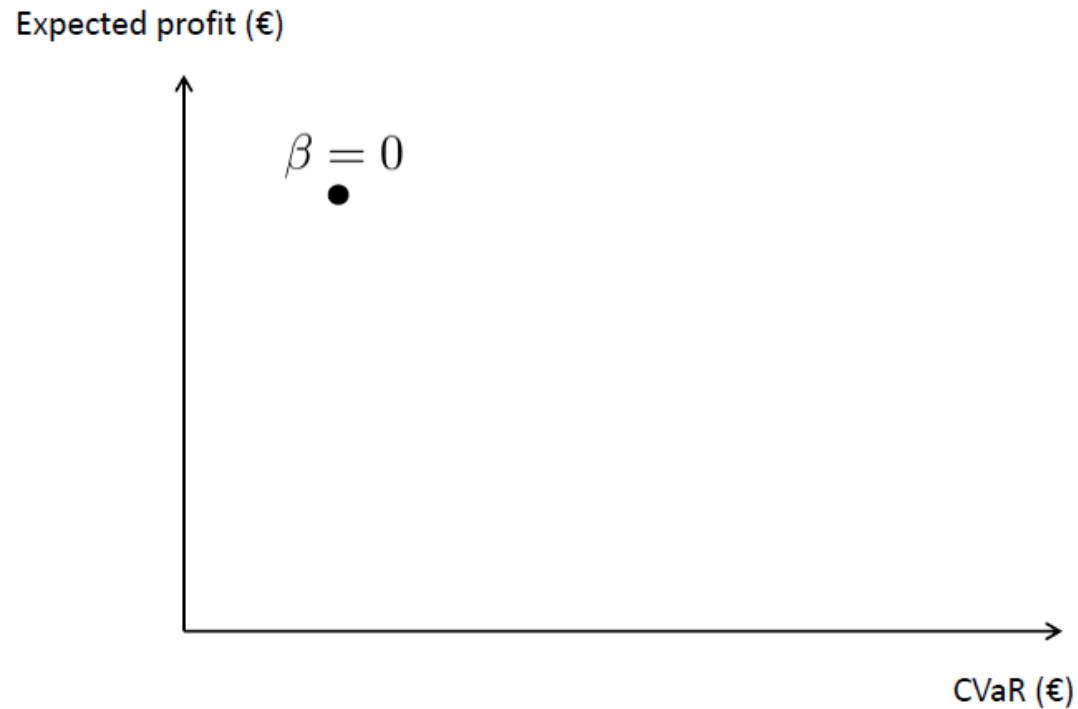
$$\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} (1 - \beta) \left[\sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \right].$$

Both versions of the weighted objective function can be used.

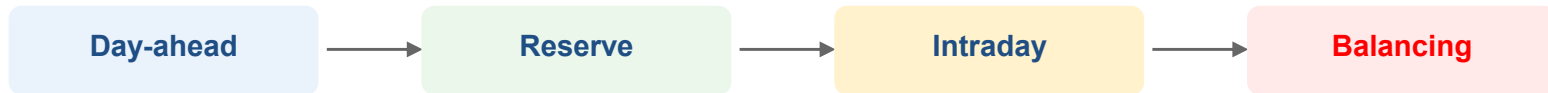
In both, $\beta = 0$ implies a **risk-neutral** case (expected profit maximization only).

In both, higher values of β imply greater conservativeness.

Efficient frontier



Extension: multi-market bidding portfolio



$$\text{portfolio_profit}[\omega] = \text{revenue_DA}[\omega] + \text{revenue_reserve}[\omega] + \text{revenue_intraday}[\omega] + \text{balancing_settlement}[\omega] - \text{penalties}[\omega]$$

Apply CVaR to $\text{portfolio_profit}[\omega]$,
not to each market separately.

Coupled decisions

Capacity sold in one market reduces availability in another.

Obligation risk

Reserve and DA commitments create delivery obligations.

Portfolio view

CVaR rewards hedging between markets and technologies.

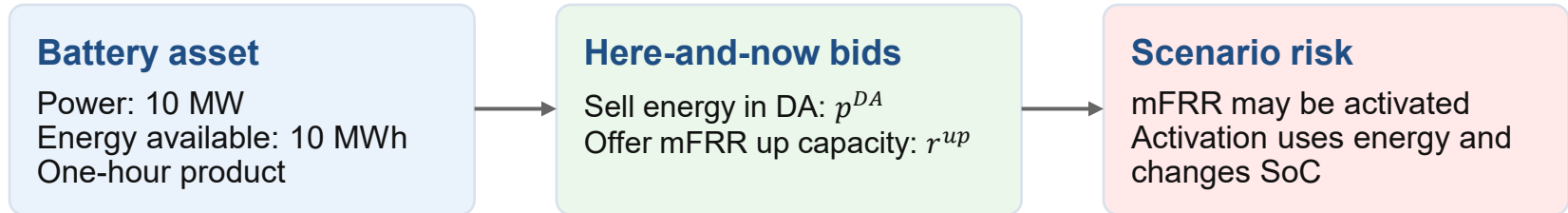
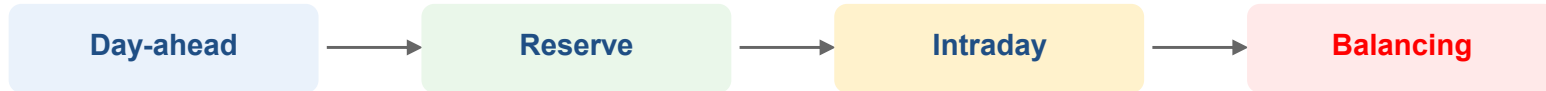
Risk-averse price-taking offering model

New additions (all other parts remain unchanged):

$$\begin{aligned}
 & \max_{p_t^{\text{DA}}, \delta_{t,\omega}^+, \delta_{t,\omega}^-, \zeta, \eta_\omega} \quad (1 - \beta) \sum_{\omega \in \Omega} \pi_\omega \text{Profit}_\omega + \beta \left(\zeta - \frac{1}{1 - \alpha} \sum_{\omega \in \Omega} \pi_\omega \eta_\omega \right) \\
 & \text{s.t.} \quad 0 \leq p_t^{\text{DA}} \leq P^{\text{nom}}, \quad \forall t \in \mathcal{T}, \\
 & \quad \delta_{t,\omega} = p_{t,\omega}^{\text{real}} - p_t^{\text{DA}}, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\
 & \quad \delta_{t,\omega} = \delta_{t,\omega}^+ - \delta_{t,\omega}^-, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\
 & \quad \delta_{t,\omega}^+ \geq 0, \quad \delta_{t,\omega}^- \geq 0, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\
 & \quad \text{Profit}_\omega = \sum_{t \in \mathcal{T}} (\lambda_{t,\omega}^{\text{DA}} p_t^{\text{DA}} + \lambda_{t,\omega}^{\text{up}} \delta_{t,\omega}^+ - \lambda_{t,\omega}^{\text{down}} \delta_{t,\omega}^-), \quad \forall \omega \in \Omega, \\
 & \quad \eta_\omega \geq \zeta - \text{Profit}_\omega, \quad \forall \omega \in \Omega, \\
 & \quad \eta_\omega \geq 0, \quad \forall \omega \in \Omega.
 \end{aligned}$$

Here, π_ω denotes the probability of scenario ω , while $\beta \in [0, 1]$ determines the trade-off between expected profit and CVaR-based tail protection.

Battery example: bidding in DA + mFRR



Feasibility for upward mFRR:

$$p^{DA} + r^{up} \leq p^{\max}$$

$$p^{DA} + a_{\omega} r^{up} \leq E^{\text{available}} \quad \forall \omega \in \Omega$$

where a_{ω} is the activation fraction.

Scenario profit:

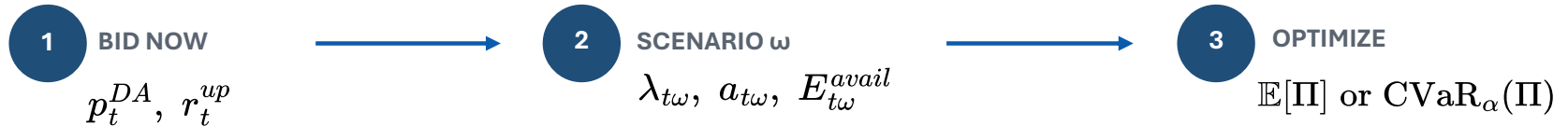
$$\Pi_{\omega} = \lambda^{DA} p^{DA} + \lambda^{\text{cap}} r^{up} + \lambda_{\omega}^{\text{act}} a_{\omega} r^{up} - \gamma_{\omega}, \quad \forall \omega \in \Omega$$

Where γ_{ω} represents the penalty of not satisfying allocated capacity in scenario ω .

The same first-stage bid must remain feasible and profitable under uncertain mFRR activation.

Battery DA + mFRR: two-stage stochastic formulation

First-stage bids are fixed before uncertainty; scenario prices, activation and availability determine realized profit and tail risk.



1 First-stage bids and physical coupling

Decision variables

p_t^{DA} day-ahead discharge bid
 r_t^{up} upward mFRR capacity bid

Scenario-dependent states

$a_{t,\omega}$ activation fraction
 $e_{t,\omega}$ state of charge
 $\Pi_{t,\omega}$ realized portfolio profit

Physical limits

$$p_t^{DA} + r_t^{up} \leq P^{\max}$$

$$p_t^{DA} + a_{t,\omega} r_t^{up} \leq E_{t,\omega}^{avail}$$

2 Scenario profit and risk objective

Scenario profit

$$\Pi_\omega = \sum_t [\lambda_{t,\omega}^{DA} p_t^{DA} + \lambda_{t,\omega}^{cap} r_t^{up} + \lambda_{t,\omega}^{act} a_{t,\omega} r_t^{up} - \gamma_{t,\omega}]$$

Risk-neutral objective

$$\max \sum_{\omega \in \Omega} \pi_\omega \Pi_\omega$$

Risk-averse objective

$$\max (1 - \beta) \mathbb{E}[\Pi] + \beta \text{CVaR}_\alpha(\Pi)$$

mFRR capacity is attractive only when activation profit and energy feasibility remain robust across scenarios.

Battery example: energy-only vs DA + mFRR

Assume $\lambda^{\text{DA}} = 85 \text{ €/MWh}$, $\lambda^{\text{cap}} = 50 \text{ €/MW}$, activation price = 120 €/MWh, one-hour product.

Bid	p^{DA}	r^{up}	No activation	50% activation	100% activation	Expected profit	Worst outcome
A: DA only	10	0	850 €	850 €	850 €	850 €	850 €
B: DA + mFRR	5	5	675 €	975 €	1275 €	975 €	675 €

Risk-neutral view

Bid B has the highest expected profit because it earns capacity and activation revenue.

Risk-averse view

Bid A has a better worst outcome and higher CVaR for the lower tail.

Key constraint

Avoid $p^{\text{DA}} + r^{\text{up}} > 10 \text{ MW/MWh}$ in full activation.

Multi-market bidding is not only revenue stacking; it also creates coupled delivery obligations and tail risk.

Battery example: expected profit vs tail protection

Bid	Mean	Worst	CVaR _{2/3}	Interpretation
A: DA only	850 €	850 €	850 €	stable but lower mean
B: DA + mFRR	975 €	675 €	675 €	higher mean, weaker tail

For three equiprobable scenarios, CVaR_{2/3} equals the average of the worst one-third outcomes.

$\beta = 0$

The model selects the highest expected profit: likely Bid B.

β high

The model values downside protection: Bid A may dominate.

Operational insight

mFRR participation is attractive only if tail exposure is acceptable.

Problem description: battery bidding in DA + mFRR markets

Asset & markets

- Grid-connected battery (BESS): power limits, SOC limits, charge/discharge efficiencies, degradation cost.
- Participates in the day-ahead (DA) energy market and the mFRR market (capacity bids + energy activation).
- When activated, the battery delivers up/down power toward the TSO's activation target.
- 24 h horizon: 15-min energy steps ($T = 96$), hourly reserve bids ($H = 24$).

Decisions in two stages

Stage 1 (here-and-now, $D-1$):

- DA charge/discharge schedule.
- mFRR up/down capacity bids with min-bid rules and headroom/footroom limits.

Stage 2 (recourse, per scenario ω):

- Actual charging/discharging and SOC trajectory.
- Delivered activation vs. the TSO activation target; over-/under-delivery penalized.

Uncertainty & objective

Unknown at bid time:

- DA prices, mFRR capacity & activation prices, TSO activation volumes.
- Modeled by scenarios $\omega \in \Omega$ with probabilities $\pi\omega$.

Goal:

- Maximize expected profit: DA + capacity + activation revenue – imbalance, tariff & degradation costs.
- Risk-averse variant: mean-CVaR trade-off with weight β .

Core difficulty: bids are fixed before prices and activations are realized → two-stage stochastic MILP (binaries for bid flags and charge/discharge exclusivity).

Two-stage stochastic MILP: battery in DA + mFRR markets

Sets & parameters

Sets:

$T = \{0, \dots, T-1\}$: time steps $T=96$, $\Delta t=0.25$ h

$\mathcal{H} = \{0, \dots, H-1\}$: hours $H=24$

ω : scenarios, $\Pi_\omega = 1/|\Omega|$

Battery physics:

η_{ch}, η_{dis} : efficiencies
 $P_{ch}^{max}, P_{dis}^{max}$: power limits (MW)
 E_{min}, E_{max} : SOC limits
 E_{init} : initial/terminal SOC
 $B_{min,t}$: min bid, SoC margin

Exogenous (ω -indexed):

$\lambda_{t,\omega}^{DA}$: day-ahead price
 $\lambda_{t,\omega}^{up}, \lambda_{t,\omega}^{dn}$: mFRR prices
 $\lambda_{t,\omega}^{c,up}, \lambda_{t,\omega}^{c,dn}$: cap. prices
 $V_{t,\omega}^{up}, V_{t,\omega}^{dn}$: TSO target activation (MW)

Stage 1 (here-and-now, D-1)

Variables (not scenario-indexed):

$p_t^{DA} \in [-P_{ch}^{max}, P_{dis}^{min}]$, $\forall t \in T$: DA schedule
 $r_h^{up}, r_h^{dn} \geq 0$, $\forall h \in \mathcal{H}$: reserve cap bids
 $y_h^{up}, y_h^{dn} \in \{0,1\}$, $\forall h \in \mathcal{H}$: bid flags

Constraints:

$$\begin{aligned} -P_{ch}^{max} &\leq p_t^{DA} \leq P_{dis}^{max} \\ B_{ch}^{min} y_h^{up} &\leq r_h^{up} \leq P_{dis}^{max} y_h^{up} \\ B_{ch}^{min} y_h^{dn} &\leq r_h^{dn} \leq P_{ch}^{max} y_h^{dn} \end{aligned}$$

Headroom:

$$p_t^{DA} + r_h^{up} \leq P_{dis}^{max}$$

Footroom:

$$p_t^{DA} - r_h^{dn} \geq -P_{ch}^{max}$$

No simultaneous reserve direction:

$$y_h^{up} + y_h^{dn} \leq 1, \quad \forall h \in \mathcal{H}$$

Decision timing: DA schedule and mFRR capacity bids are chosen before prices/activations are realized.

Stage 2 (recourse, per scenario ω)

Variables (scenario-indexed):

$p_{t,\omega}^{ch}, p_{t,\omega}^{dis} \geq 0$: Charging and discharging power
 $e_{t,\omega} \in [E_{min}, E_{max}]$: SOC
 $\delta_{t,\omega}^{up}, \delta_{t,\omega}^{dn} \geq 0$: Actual activation delivered by battery
 $d_{t,\omega}^+, d_{t,\omega}^- \geq 0$: Over/Under-delivery
 $u_{t,\omega} \in \{0,1\}$: No-sim. flag

SOC dynamics $\forall t, \forall \omega$:

$$e_{t+1,\omega} = e_{t,\omega} + \eta_{ch} \Delta t p_{t,\omega}^{ch} - \frac{\Delta t}{\eta_{dis}} p_{t,\omega}^{dis}$$

$$e_{0,\omega} = e_{T,\omega} = e^{init}, \quad \forall \omega \in \Omega.$$

Power link $\forall t, \forall \omega$:

$$p_{t,\omega}^{dis} - p_{t,\omega}^{ch} = p_t^{DA} + V_{t,\omega}^{up} - V_{t,\omega}^{dn} + d_{t,\omega}^+ - d_{t,\omega}^-$$

$$d_{t,\omega}^+ > 0: \text{ over-delivery relative to the target,}$$

$$d_{t,\omega}^- > 0: \text{ under-delivery relative to the target.}$$

No simultaneous charge/discharge:

$$p_{t,\omega}^{ch} \leq P_{max}^{ch} (1 - u_{t,\omega}),$$

$$p_{t,\omega}^{dis} \leq P_{max}^{dis} u_{t,\omega}.$$

SoC adequacy for reserve bids:

$$e_{t,\omega} - E_{min} \geq \frac{\tau}{\eta_{dis}} r_{h(t)}^{up}, \text{ [energy available for discharge],}$$

$$E_{max} - e_{t,\omega} \geq \eta_{ch} \tau r_{h(t)}^{dn}, \text{ [capacity available to absorb energy].}$$

Activation deliverability $\forall t, \forall \omega$:

$$0 \leq V_{t,\omega}^{up} + d_{t,\omega}^+ - d_{t,\omega}^- \leq r_{h(t)}^{up},$$

$$0 \leq V_{t,\omega}^{dn} - d_{t,\omega}^+ + d_{t,\omega}^- \leq r_{h(t)}^{dn}.$$

Recourse variables adapt charging/discharging and imbalance deviations after scenario realization.

Objective (risk-neutral):

$$\max \sum_{\omega \in \Omega} \pi_\omega \left\{ \Delta t \sum_t \lambda_{t,\omega}^{DA} p_t^{DA} + \sum_h (\lambda_{h,\omega}^{c,up} r_h^{up} + \lambda_{h,\omega}^{c,dn} r_h^{dn}) + \right. \\ \left. \Delta t \sum_t (\lambda_{t,\omega}^{up} V_{t,\omega}^{up} - \lambda_{t,\omega}^{dn} V_{t,\omega}^{dn}) + \Delta t \sum_t (\lambda_{t,\omega}^{up} d_{t,\omega}^+ - \lambda_{t,\omega}^{dn} d_{t,\omega}^-) - \Delta t \sum_t [c^{tariff} p_{t,\omega}^{ch} + c^{deg} (p_{t,\omega}^{ch} + p_{t,\omega}^{dis})] \right\}$$

CVaR extension: mean–CVaR objective for risk-averse bidding

Scenario profit (new variable)

$$\text{Profit}(\omega) = \text{DA_rev}(\omega) + \text{Cap_rev}(\omega) + \text{Act_rev}(\omega) + \text{Dev_rev}(\omega) - \text{Cost}(\omega)$$

$$\text{DA_rev}(\omega) = \Delta \sum_t \pi_{t,\omega}^{\text{DA}} p_t^{\text{DA}}$$

$$\text{Cap_rev}(\omega) = \sum_h (\pi_{h,\omega}^{\text{c,up}} r_h^{\text{up}} + \pi_{h,\omega}^{\text{c,dn}} r_h^{\text{dn}})$$

$$\text{Act_rev}(\omega) \& = \Delta t \sum_t (\lambda_{t,\omega}^{\text{up}} V_{t,\omega}^{\text{up}} - \lambda_{t,\omega}^{\text{dn}} V_{t,\omega}^{\text{dn}}) + \Delta t \sum_t (\lambda_{t,\omega}^{\text{up}} d_{t,\omega}^+ - \lambda_{t,\omega}^{\text{dn}} d_{t,\omega}^-)$$

$$\text{Cost}(\omega) \& = \Delta t \sum_t [c^{\text{tariff}} p_{t,\omega}^{\text{ch}} + c^{\text{deg}} (p_{t,\omega}^{\text{ch}} + p_{t,\omega}^{\text{dis}})]$$

CVaR reformulation (Rockafellar & Uryasev, 2000)

Auxiliary variables:

$\zeta \in \mathbb{R}$ (free variable) $\leftarrow$ VaR level
 $\xi_\omega \geq 0, \quad \forall \omega \in \Omega$ $\leftarrow$ shortfall below VaR

Shortfall constraint $\forall \omega \in \Omega$:

$$\xi_\omega \geq \zeta - \text{Profit}_\omega,$$

$$\xi_\omega \geq 0, \quad \forall \omega \in \Omega.$$

CVaR $_\alpha$ expression:

$$\text{CVaR}_\alpha = \zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \xi_\omega$$

Mean–CVaR objective

$$\max \quad (1-\beta) \sum_{\omega \in \Omega} \pi_\omega \text{Profit}_\omega + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \xi_\omega \right]$$

Subject to:

- All stage-1 and stage-2 constraints
- Profit definition (equality constraints)
- CVaR shortfall constraints

Risk parameters:

$\alpha \in (0,1)$ confidence level (typical values: 0.90, 0.95, 0.99)

CVaR $_\alpha$ = avg profit of worst $(1-\alpha)$ fraction

$\beta \in [0,1]$ risk-aversion weight

$\beta = 0 \rightarrow$ risk-neutral: $\max E[\text{Profit}]$

$\beta = 1 \rightarrow$ fully risk-averse: $\max \text{CVaR}_\alpha$

$0 < \beta < 1 \rightarrow$ mean–CVaR trade-off

Additions to risk-neutral model:

- +1 free variable (ζ)
- + $|\Omega|$ variables ($\xi_\omega \geq 0$)
- + $|\Omega|$ constraints (shortfall)
- + $|\Omega|$ constraints (Profit definition)

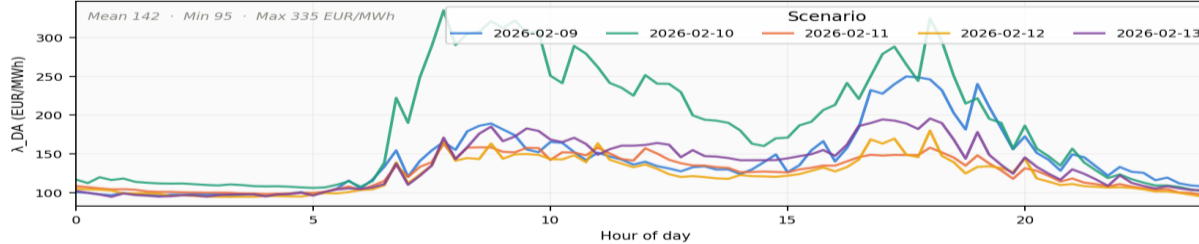
same MILP solver applies.

$\beta > 0$: same stage-1/stage-2 constraints + two new variable blocks + one new objective term.

NO3 Input data overview

NO3 area, 5 scenarios (09–13 Feb 2026) · DA · mFRR activation · reserve capacity prices · net activated volume

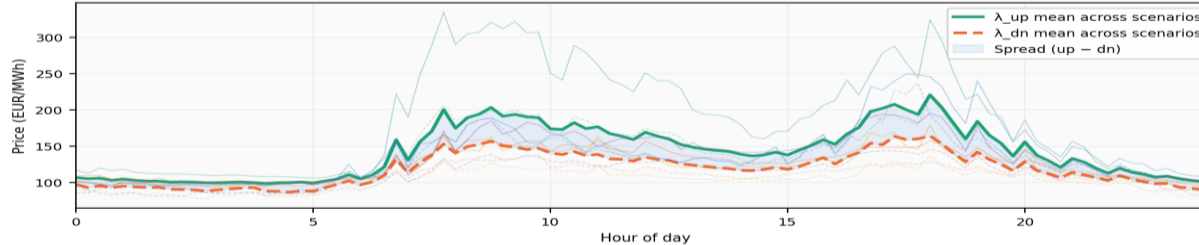
(A) Day-ahead prices — 15-min resolution



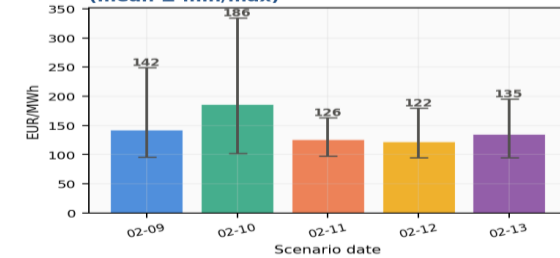
(B) Dataset summary

Scenarios	5 days (Feb 2026)
Resolution DA	15-min (T = 96 steps)
Res. cap.	hourly (H = 24 steps)
Time step Δt	0.25 h
DA price range	95 – 335 EUR/MWh
DA price mean	142.5 EUR/MWh
mFRR up range	95 – 335 EUR/MWh
mFRR dn range	78 – 237 EUR/MWh
Cap up range	1.00 – 25.70 EUR/MW
Cap dn range	1.93 – 4.10 EUR/MW
NAV range	-683 – 111 MW
NAV mean	-136.4 MW (net downward)

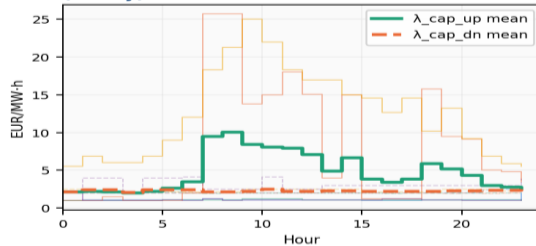
(C) mFRR activation prices — 15-min resolution



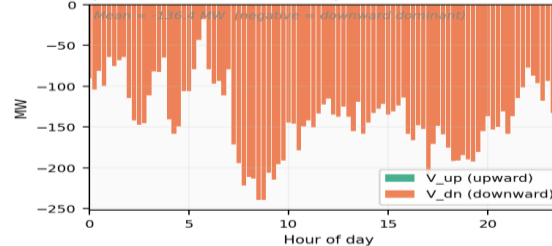
(D) DA price per scenario (mean ± min/max)



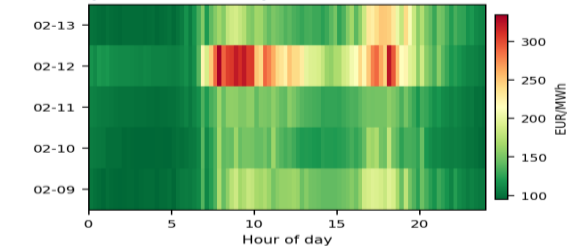
(E) Capacity prices (hourly)



(F) Net activated volume (mean across scenarios)



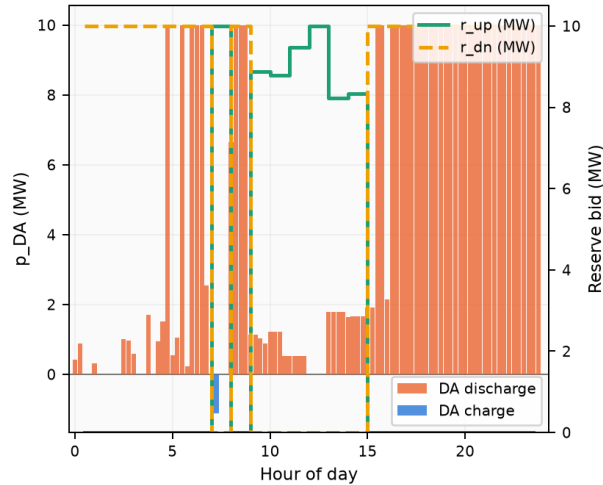
(G) DA price heatmap (scenario × hour)



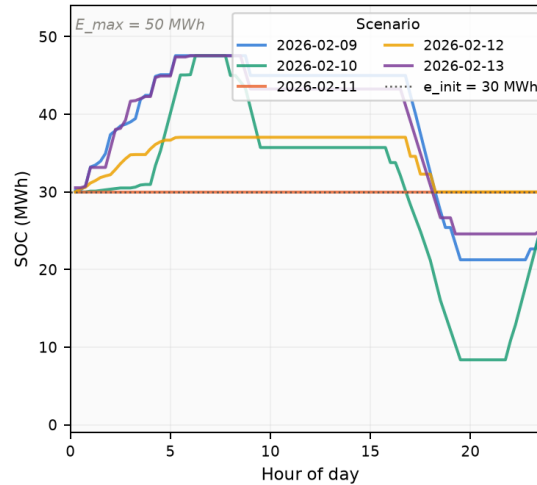
Battery in DA+mFRR multi Market

10 MW/50 MWh, 5 Scenarios, $\alpha = 0.95$ $\beta = 0.5$

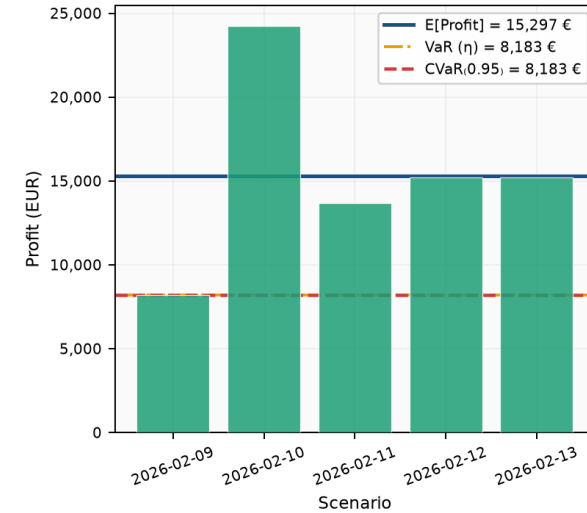
(A) Stage-1 bids



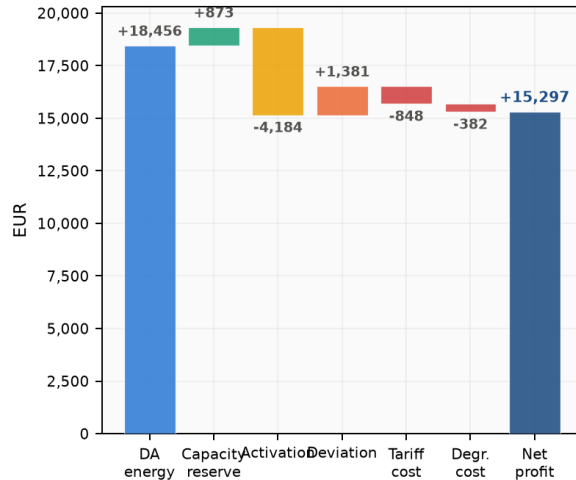
(B) Battery SOC trajectories



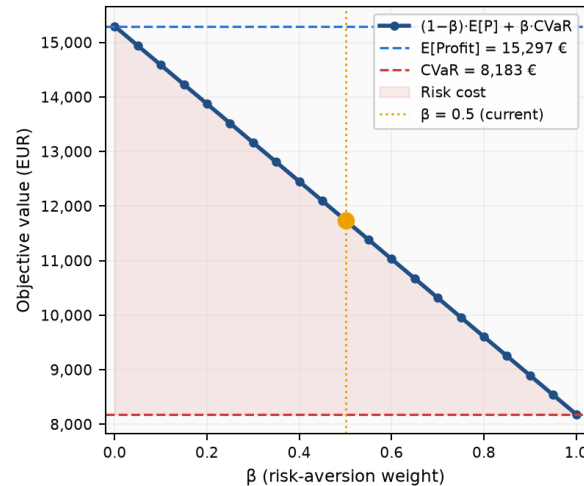
(C) Scenario profit distribution



(D) Revenue stacking (mean, EUR)



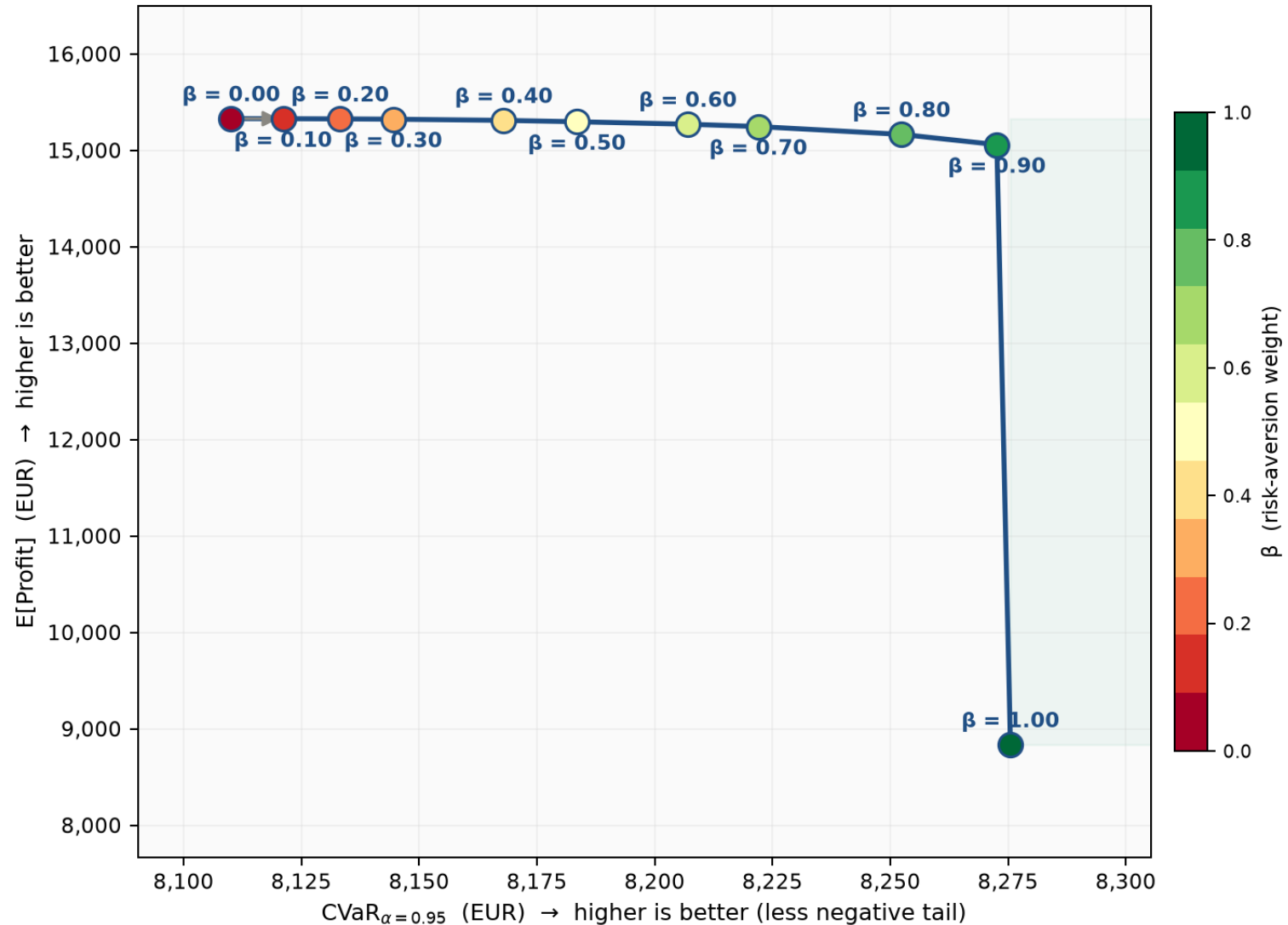
(E) β sensitivity (mean-CVaR trade-off)



(F) Model summary & risk measures

Battery	10 MW / 50 MWh
Scenarios	5 ($\pi_w = 0.200$)
α (confidence)	0.95 $\rightarrow$ worst 5%
β (risk weight)	0.50 $\rightarrow$ risk-averse
$E[\text{Profit}]$	+15,297.0 EUR
$\text{VaR}(\eta)$	+8,183.5 EUR
$\text{CVaR}(0.95)$	+8,183.5 EUR
Objective	+11,740.2 EUR
Best scenario	+24,237.7 EUR
Worst scenario	+8,183.5 EUR
Max r_{up} bid	10.00 MW ($h=7$)
Max r_{dn} bid	10.00 MW ($h=0$)

Efficient frontier (expected profit vs CVaR)



What changes when the bidder becomes risk-averse?

Risk-neutral

May offer close to expected production if expected settlement is favorable.

higher expected profit

Risk-averse

May withhold volume to avoid expensive shortage in low-wind scenarios.

higher tail profit

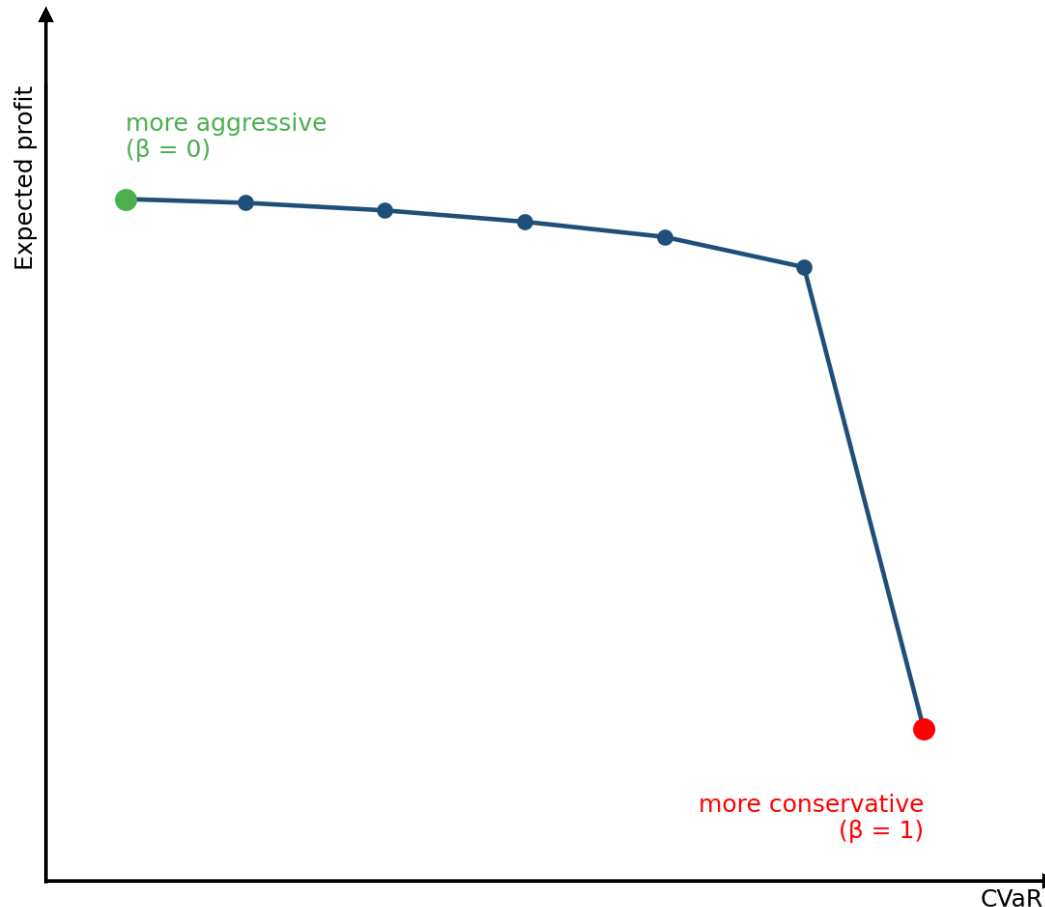
Multi-market

May allocate capacity to reserves or intraday as a hedge.

diversified revenue

Risk aversion does not mean “bid less” in every hour; it means “bid differently where the tail is dangerous”.

Trade-off: expected profit vs. CVaR



β sweep

Solve the model for $\beta = 0, 0.25, 0.5, 0.75, 1$.

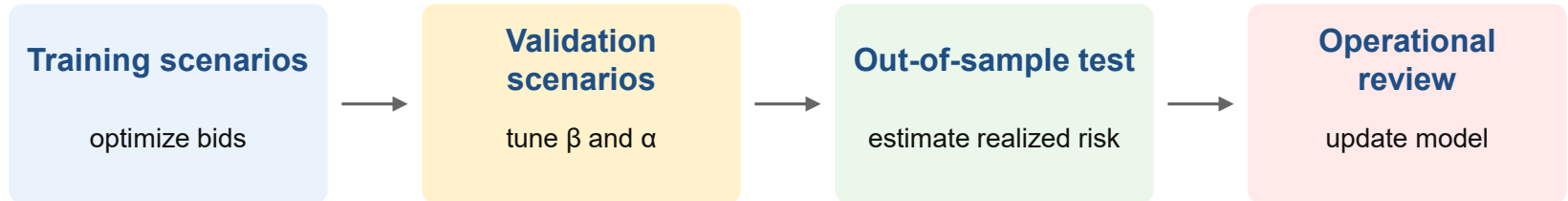
Interpretation

The efficient frontier shows the cost of risk reduction.

Decision

Pick β based on risk appetite or mandated risk limit.

Testing the quality of risk-aware decisions



In-sample danger

A decision can look excellent on the scenarios used for optimization.

Out-of-sample discipline

Evaluate on unseen historical days or simulated scenarios.

Practical modelling choices

- Choice of α : common values are 0.90, 0.95, 0.99 depending on risk policy.
- Choice of β : sensitivity analysis is more transparent than selecting one hidden number.
- Scenario quality matters: biased tails lead to misleading CVaR decisions.
- Settlement design matters: one-price and two-price balancing change incentives.
- Portfolio scope matters: risk should be measured over the actor's real cash-flow boundary.

Key takeaways

- Risk management is about the entire profit distribution, not only its average.
- VaR identifies the tail threshold; CVaR measures the average outcome in the bad tail.
- CVaR has a linear reformulation for scenario-based stochastic programming.
- A mean-CVaR objective provides a transparent trade-off between profit and downside protection.
- In multi-market bidding, risk should be evaluated on portfolio-level cash flow.